IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Digital Object Identifier

Physics-Informed Bayesian Event Attribution From Sparse PMUs

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]

[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

Corresponding author: Jorge Luis Mayorga Taborda (e-mail: jl.mayorga236@uniandes.edu.co).

ABSTRACT Sparse PMU measurements can reveal a disturbance without uniquely identifying its cause. We study how network equations and conditional Bayesian evidence complement feature-based classification for this task. A bank of switched differential-algebraic predictors compares source, onset, severity and restoration histories, retaining electrical and instrumental memory between interventions. Local severity integration and fixed-network model averaging produce separate active-event and originating-cause masses. Component experiments examine hidden-voltage reconstruction, engineered-feature localization and mixed-source response sensitivities; these are distinguished from the final causal estimator. In a frozen IEEE 39-bus comparison with eight PMUs, 18 six-event schedules are replayed under three network conditions and two noise levels. The physical bank makes 644/648 timely family/source decisions with 72 extra alarms, compared with 330/648 and 29 for a causal ExtraTrees reference. Its strongest gain is load-source attribution; line attribution is similar to the short-window learned comparator. Most extra physical alarms occur under network changes outside the model bank. A separate historical transfer experiment retains the original hackathon submission and its confusion matrix. The results identify a useful coverage–alarm trade-off within the simulated catalog. Transfer to a different network model, calibration of event probabilities and the separate contribution of each inference component require further evidence.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

EIGHT PMUs observe only a small part of a 39-bus network, yet an operator needs to know whether a disturbance occurred, which physical mechanism caused it, and where it began. An unmeasured bus can change several remote PMU signals, while different sources can produce similar measured responses. The problem is therefore both electrical and statistical: reconstruct the response allowed by the network, then compare its possible causes under uncertain state, severity and noise.

Our starting point is a feature-based hackathon system. Voltage, current, frequency and spatial/temporal descriptors supported an ExtraTrees detector, classifier and localizer. We retain that route and its results. The complementary route asks an explicit counterfactual question: if this source had changed at a candidate past time, what would all eight PMUs have measured up to now? Physical equations provide the candidate path; conditional Bayesian evidence accounts for unknown severity and competing histories. After restoration, those histories must still explain the remaining machine motion.

Physical restoration does not erase machine motion. A flat reference can mislabel its remaining oscillation as a second event; suppressing all alarms during recovery can hide an actual second event. A corrected voltage offset also need not correct the response to a later input.

DAE estimators recover unmeasured electrical states [1]; outage detectors compare physical hypotheses [2], [3]; and simulated trajectories support sparse-PMU fault diagnosis [4]. These ingredients are established. Our question is whether a continuing recovery should explain a new PMU change or yield to a later intervention hypothesis.

We formulate a causal attribution procedure that keeps the estimated physical response after an intervention ends and tests subsequent changes against that continuing response. The paper makes three contributions:

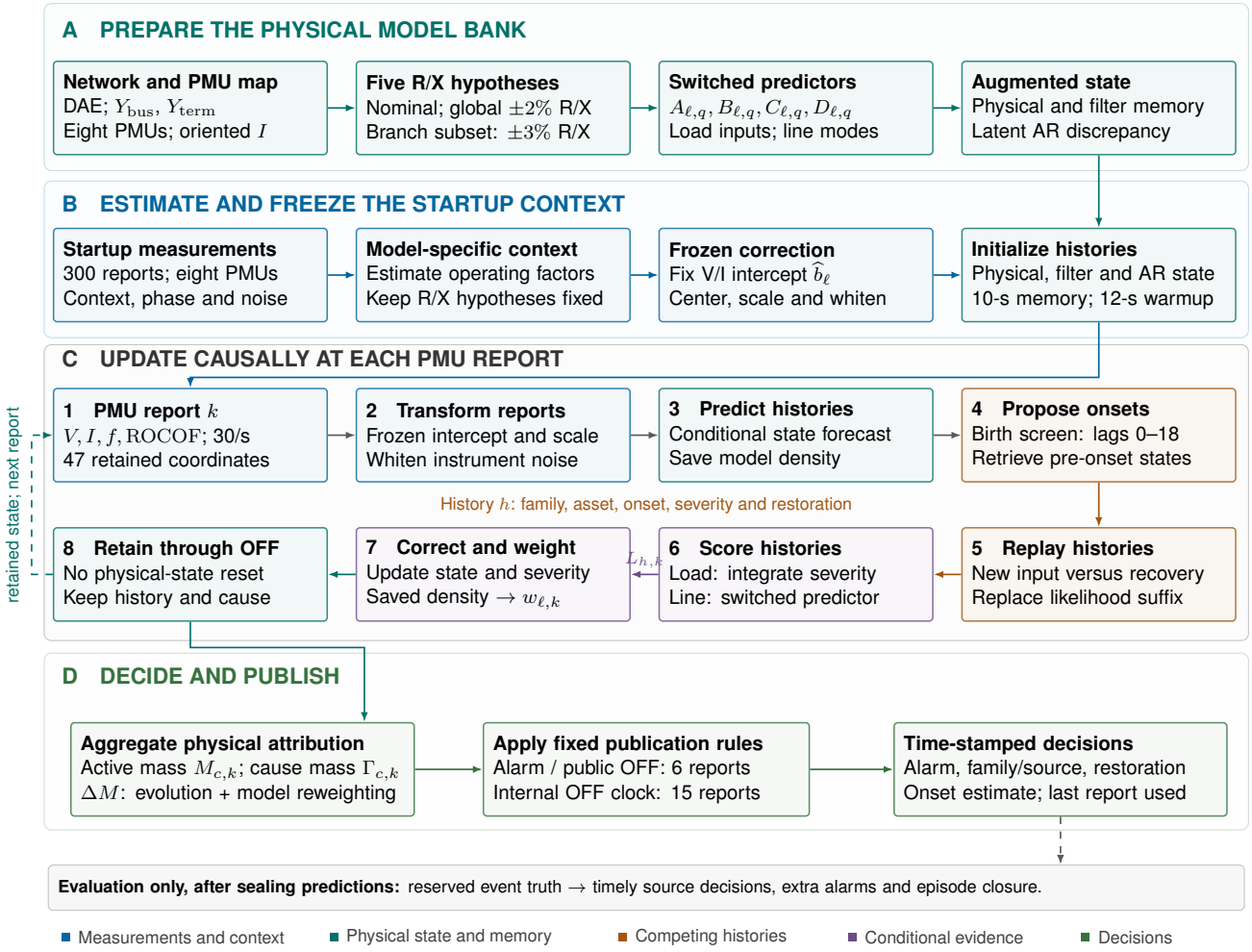


FIGURE 1. Implemented physical inference in four stages. (A) Construct switched network and PMU predictors. (B) Estimate the startup context and fix its observation correction. (C) Compare new intervention histories with retained recovery at each report, using stored pre-onset states. (D) Aggregate attribution and publish qualified alarm, source and restoration decisions. The public six-report rules and the internal 15-report OFF clock are distinct; their support conditions are specified in Section IV. Dashed feedback retains state for the next report. Reserved truth enters only the later evaluation.

1) Network-conditioned state reconstruction, physical feature ablations and mixed-source sensitivity experiments measure how network equations recover unmeasured voltages and distinguish candidate inputs. Each experiment retains its own data and evaluation conditions.
2) The continuous procedure compares new interventions with retained recovery histories, integrates local load severity, and separates active mass M from originating-cause mass Γ. Its ΔM records the changing attribution. These quantities summarize the hypothesis weights; the physical information enters through the candidate trajectories.
3) A direct historical transfer of the submitted system and a new same-stream comparison with causal ExtraTrees distinguish feature failure, source attribution and alarm cost. The latter also compares a published outage detector on its eligible persistent-line task.

The complete physical bank attributes 322/324 load realizations before OFF versus 15/324 for the causal ExtraTrees reference. Under out-of-bank R/X it raises 68 versus 10 unmatched alarms. Thus the supported gain is timely load attribution with a larger alarm burden. A subsequent controlled study separates amplitude-evidence scoring from physical-state carry at closure. Its design and completion status are reported separately from the original comparison.

The test uses nine candidate load buses and nine lines in a fixed IEEE-39 simulation. It does not establish an all-family hackathon score, calibrated event probabilities or real-time deployment.

II. RELATED WORK

DAE and unknown-input estimators recover hidden states or inputs [1], [5], [6]; functional observability and diagnosability ask which states or sources can be distinguished [7]–[10]. Physical outage inference already uses network hypotheses, uncertain states and sequential tests [2], [3], [11], [12]. Sparse input recovery estimates forcing locations and unknown ini-

tial states in linear dynamics [13]. Sensor placement limits the information available at remote PMUs [14]–[16].

Warner *et al.* compare simulated voltage/current paths across an unknown-onset window for short-circuit line identification [4]. Our load/line task continues through restoration and successive events; their method has not been run on this cohort. Multiple-model filters and adaptive history banks already retain conditional modes [17], [18]; sequential diagnosis has established error/delay theory under specified observation laws [19]. Our screened, pruned bank inherits none of those general optimality or delay guarantees.

Learned PMU classifiers and physical uncertainty models provide other routes [20]–[26]. For example, Li *et al.* embed disturbance-type/topology and localization-cost constraints in a spatiotemporal graph-convolutional model [21]. Our physical route instead computes the observation likelihood of candidate interventions; its graph is the admittance network and terminal map, not a learned message-passing layer. The two frozen ExtraTrees implementations test supervised references, not all learned diagnosis. The new contribution claimed here is the specified sparse-PMU procedure and its evidence: new interventions and retained recovery histories compete for the same causal reports, with component experiments explaining the role and limits of physical representation. We do not claim that combining physics, Bayes and a tensor is itself new. Table 1 distinguishes the neighboring tasks and the comparators actually executed here.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$y_k = \mathcal{M}_k(x_{0:k}, z_{0:k}, q_{0:k}; \theta_M) + b + \varepsilon_k. \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. The causal measurement map uses current algebraic variables for voltage and terminal current and retained phase/instrument history for frequency and ROCOF. The history notation makes that memory explicit; the instantaneous electrical map used in the Schur reduction below is its voltage/current component. A machine's rotor speed is not substituted for bus frequency.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 gives the report-level logic. A history is a proposed cause and trajectory, not just a bus label. The inference bank predicts the next report under no new event, a new load or line action, or retained recovery. Its switched predictors come from the network DAE and oriented PMU terminal map. Statistical filtering handles uncertain states and residual error. The nonlinear simulator generates the observations; inference uses local predictors, not a new full simulation for every hypothesis and report.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{\mathrm{bus}} = Y_{\mathrm{bus}} V, \quad I_{\mathrm{term},r} = Y_{\mathrm{term},r} V, \quad S_i = V_i I^*_{\mathrm{bus},i}. \tag{5}$$

TABLE 1. Closest inference contracts. Chen and the two fixed ExtraTrees adapters are the executed comparators. An untested capability is not an algorithmic impossibility.

Route	Mechanism	Evaluated target	Role in this study
Chen / Rovatsos	Network likelihoods and sequential tests	Line outages; Rovatsos includes transients	Chen on persistent-line first decisions
Warner high-fidelity	Simulated V/I paths and onset search	Short-circuit line candidates	Closest trajectory precedent; no common-data run
Adaptive Matrix CuSum	Sequential diagnosis with error/delay analysis	One unknown distributional change	Statistical precedent; no inherited guarantee
DAE / stochastic-state estimation	Physical state and uncertainty models	States, unknown inputs or parameters	Components and adjacent tasks
Learned PMU diagnosis	Features or constrained representations	Event families and locations	Two frozen ExtraTrees references
Physical history bank here	Estimated-past-state paths and recovery	Successive catalog load/line events	Timely attribution, extras and restoration

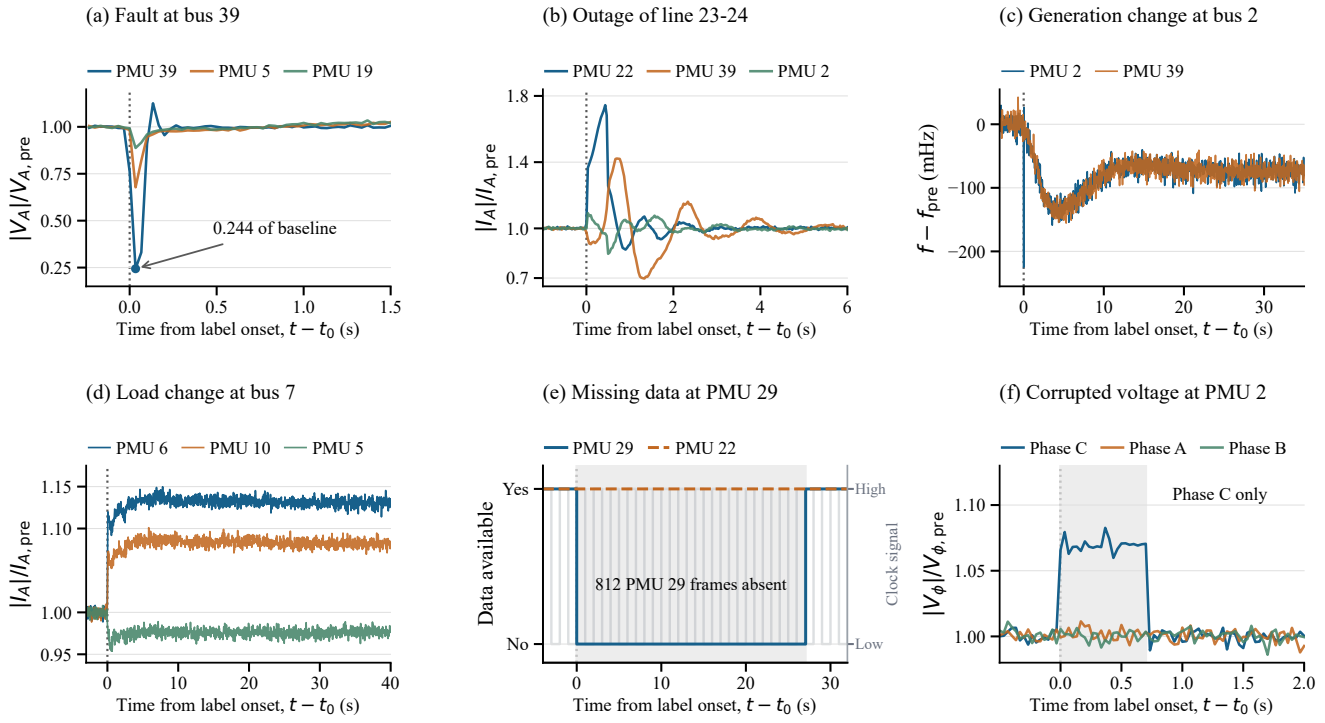


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes the branch stamp and therefore the algebraic voltages and measured currents. A terminal map changes directly only if the switched branch is itself measured; none of the nine tested line candidates is a measured PMU terminal. The load operator scales both P_{set} and Q_{set} by $1 + a$ with the fixed constant-impedance ZIP fractions, while the line operator switches the selected branch between active and open modes.

The executed plant has 114 differential states, 78 algebraic variables and 38 load-input coordinates. Its ten Sauer–Pai machines use AVR Type I with quadratic ceiling and TGOV1 governors, except the bus-39 machine, which has no external governor or excitation-control inputs. All 19 load models use constant-impedance ZIP fractions. With injection sign convention,

$$P_i = (1+a_i)P_i^0\,|V_i/V_{set,i}|^2\,, \qquad Q_i = (1+a_i)Q_i^0\,|V_i/V_{set,i}|^2\,. \tag{6}$$

Here $P_i^0 < 0$ represents consumption; reactive injection can be positive for a capacitive load. The archived equilibrium and voltage setpoints, including those at buses 31 and 39, define the executed operating point.

On a regular index-one DAE chart with a fixed angle

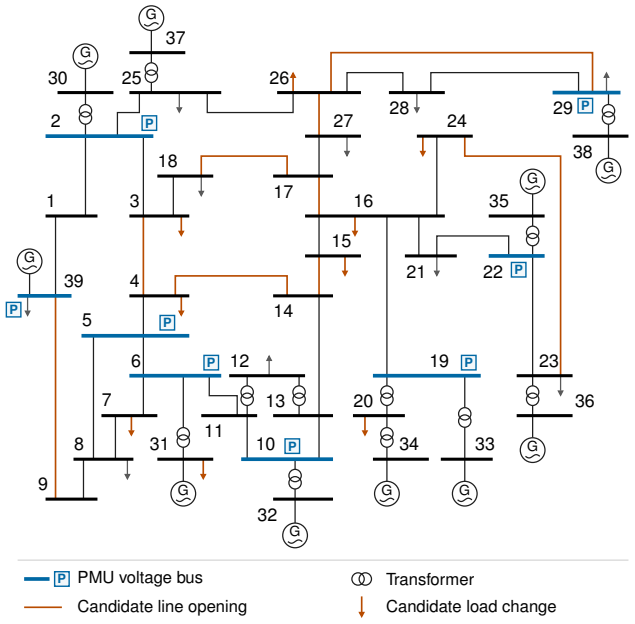


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.**

TABLE 2. **Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.**

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside causal test
Unknown	Unresolved diagnosis	Not a new physical family

reference,

$$\begin{aligned}\delta z &= -g_z^{-1}(g_x\delta x + g_u\delta u),\\ \delta\dot{x} &= A_q^{red}\delta x + B_q^{red}\delta u,\\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{7}$$

For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{8}$$

They propagate a perturbation at an unmeasured source to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in g_q and, where applicable, H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator would also contribute directly through H_q; the tested line modes keep the same terminal map. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. The generator estimates phase derivatives with a robust, magnitude-weighted quadratic fit over 21 reports. The predictor approximates that operation with a fixed unweighted quadratic fit over the same history; these operators need not agree when weights change. Inertia, controls and load law remain those of the fixed plant.

After augmenting instrumental memory, a conditional predictor has the local form

$$\begin{aligned}\xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k,\\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{9}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. Equation (9) states the generic local model; the implemented update combines propagation and observation into an operator on the previous state. The supplement gives that indexing and the associated transition gain explicitly. States from different equilibria are not interpreted as an absolute rotor-state mixture.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{10}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its frozen birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset,

onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{11}$$

The supplement derives the closed form by completing the square and works through a numerical example. It also specifies the conditional state representation $m_h + d_h a$ and its update. The local amplitude integral is exact for the stated conditional Gaussian model. The complete bank remains approximate: regime covariances are shared from nominal-operator Riccati equations, context estimates enter by substitution, and histories are screened and pruned. At episode closure the augmented state mixture mean is retained and omitted between-history spread is recorded. A fully Bayesian posterior over states, context and all histories is not claimed.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{12}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t, \kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t, \kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{13}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a trajectory, not a zero residual; it is not a new propagation theorem.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{14}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\begin{aligned} \Delta M_{c,k} = &\sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ &+ \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \end{aligned} \tag{15}$$

The terms distinguish within-model evolution from model reweighting. This accounting identity aids interpretation, without constituting a new tensor factorization. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The internal observer proposes an alarm after six reports above the 0.99 active-mass threshold. Public alarm publication additionally requires an internal pending episode and six reports of qualified active support: the frozen source screen qualifies load cells, while admitted line cells qualify through their switched operator. Public OFF requires six reports above 0.99 for the diagnosed cell's joint cause-and-restoration mass. The observer also has a separate, more conservative 15-report internal OFF clock. Diagnosis selects the family with greatest summed cause mass and then the highest-mass source within that family. It is emitted after six reports of stable cause support or at the 15-report post-alarm deadline. The supplement defines the source screen, early line rule and episode replacement, including how internal messages are propagated publicly. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture evaluates two complete configured procedures; it does not isolate the outer network bank from publication decisions. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory ablation.

The supplement records the frozen birth/restoration hazards, severity prior and integration intervals, restoration-

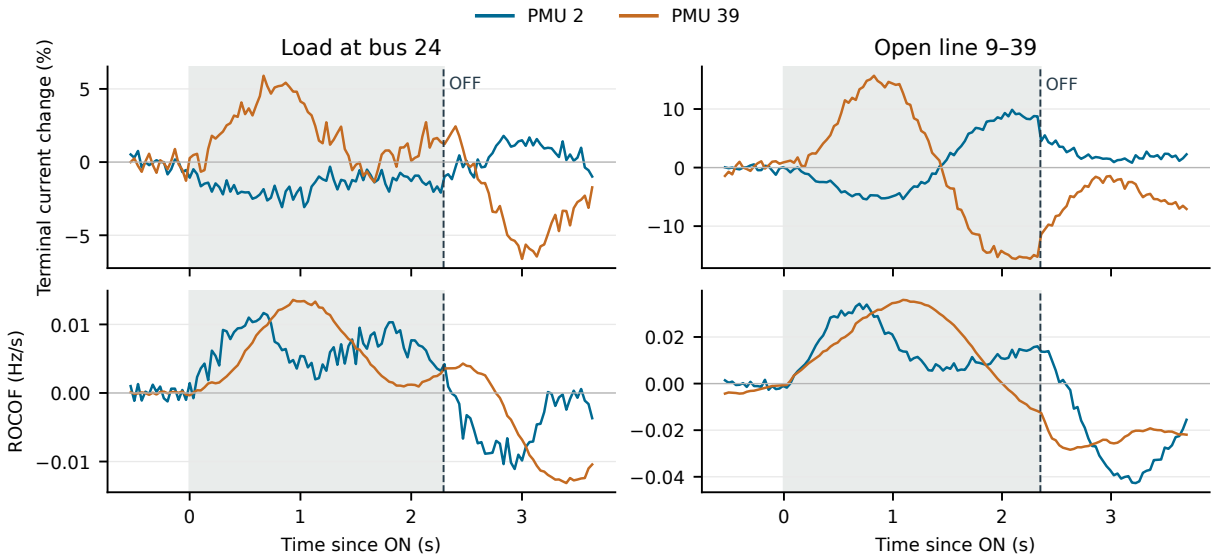


FIGURE 4. **Two interventions from one continuous development record. Current change and ROCOF at PMUs 2 and 39 remain dynamic after physical OFF (dashed line). The gray interval is the true active period, shown only for explanation; inference uses all eight PMUs without event timing. These two PMUs illustrate the traces and were fixed for both panels.**

child rules and beam limits, together with the execution sequence and illustrative calculations. These details preserve the archived configuration; the compact algorithm below states the report order and information boundary.

G. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: one causal update. Input: nominal model, fixed method configuration and PMU reports. Initialize context, reference, noise and conditional states from the 300 startup reports. Event truth is never an input.

1) Transform report k; predict it from each retained physical or recovery history. Save each network model's predictive density before screening or correcting with this report.
2) Screen onsets in the latest 18 reports. Start each proposed history from its stored pre-onset state and replay its suffix, replacing the old suffix likelihood once.
3) Score histories, integrating load severity or using the switched line operator. Then correct state and severity, update network weights, and prune by the fixed rules. Retain state and cause through restoration.
4) Aggregate active mass M, cause mass Γ and ΔM. Apply the frozen alarm, source and OFF rules; record emission time, estimated onset and last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is 30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. WHAT PHYSICAL STRUCTURE AND LEARNED FEATURES RECOVER

The continuous estimator builds on three distinct questions: what can be reconstructed at unmeasured buses, what response distinguishes candidate sources, and how evidence should be assigned when several explanations fit. The following component experiments answer parts of these questions. They use different cohorts and are not ablations of the final continuous estimator. Figure 5 places their measured effects side by side without pooling their scores.

A. RECOVERING UNMEASURED VOLTAGES IS NOT YET EVENT ATTRIBUTION

In 100 nominal test trajectories, the mean hidden-voltage relative error falls from 0.3402% for the equilibrium reference to 0.00990% for snapshot weighted least squares and 0.00963% for a causal Kalman reconstructor. Most of this improvement comes from conditioning the physical network on the measured phasors; temporal filtering adds a smaller improvement. This supports reconstruction under that model and sensor placement, not unique recovery of every source. The independently evaluated Kalman reconstructor is not silently substituted for the conditional observer in Fig. 1.

B. PHYSICAL FEATURES REMAIN A USEFUL LEARNED ROUTE

The historical feature study raises physical-source Top-1 from 49.6% with 136 features to 54.8% with graph-temporal features added, and 59.0% with the full 448-feature set. Voltage level, relative angle, frequency, time scale and spatial coherence carry complementary information; Fig. 5(b) reports an executed feature comparison rather than an inferred explanation of tree importance. Electrical features make a classifier

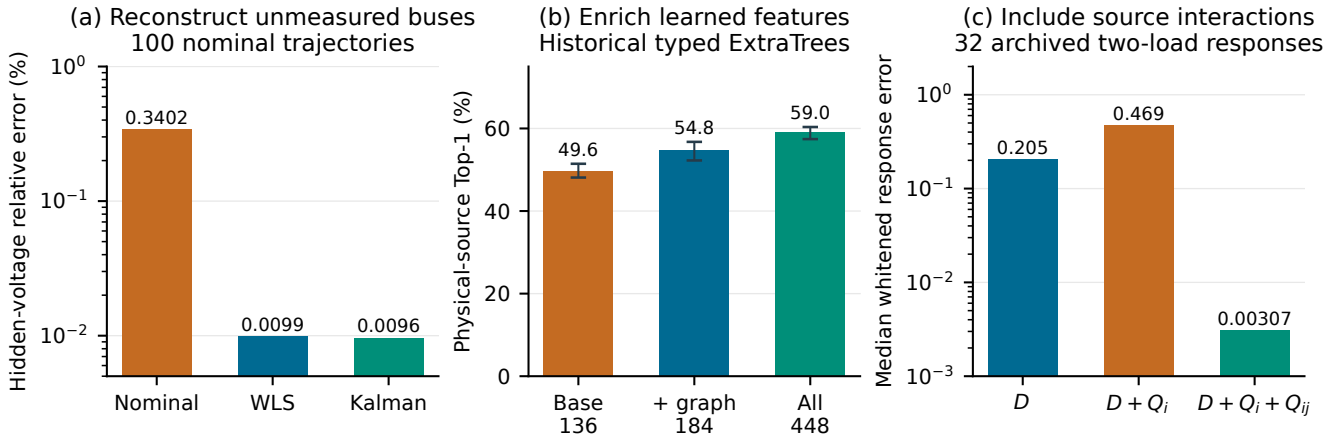


FIGURE 5. **Three separate component experiments. (a) Mean relative error of the 31 unmeasured bus voltages under a matched nominal model; this is not a PMU compliance TVE test. (b) Mean physical-source Top-1 over three model seeds, each scored on 7,915 labeled physical samples, in the historical typed-ExtraTrees feature study; whiskers show the seed range, not a confidence interval. These 136–448-feature models are neither the final hackathon submission nor the causal comparators. (c) Median whitened prediction error at the archived 120-report horizon for two-load responses: mixed-source terms matter, while adding only self-curvature worsens this metric. Each panel has its own units and cohort.**

physics-informed in its representation. They do not enforce Kirchhoff balance on its predicted label or compute the PMU response of a candidate intervention.

For phase-to-neutral RMS phasors and a declared current orientation,

$$S_{ij} = P_{ij} + \mathrm{j}Q_{ij} = \sum_{\phi\in\{a,b,c\}} V_{i\phi} I^{*}_{ij,\phi}. \tag{16}$$

This expression requires voltage and current at the same terminal. In the frozen map, the bus-2 voltage is paired with a current measured at bus 3, and the bus-10 voltage with a current at bus 11. Their raw channel products are therefore power proxies, not exact terminal powers. Even a co-located terminal power is not the bus's net injection unless all relevant terminals are accounted for. Frequency and ROCOF complement the phasors but share the instrument's angle processing; the likelihood must account for that dependence. Fundamental-frequency phasors also do not determine harmonic distortion. Thus adding nominally physical features does not by itself supply missing harmonic or nodal-injection measurements.

C. MIXED-SOURCE RESPONSE COEFFICIENTS

For simultaneous load changes, a second-order response approximation includes products of the input amplitudes. In the archived two-load experiment,

$$s_{q,n}(a) = \sum_i D_{qni} a_i + \sum_i Q_{qni} a_i^2 + \sum_{i<j} Q_{qnij} a_i a_j, \tag{17}$$

where q indexes PMU coordinates, n elapsed reports and i,j load sources. Centered nonlinear simulations estimate the coefficients; the Q terms include their Taylor scaling. The mixed term represents the interaction of two simultaneous load changes. Across 32 archived two-source responses, median whitened error is 0.205 for D, 0.469 after self-curvature, and 0.00307 after mixed terms. This is evidence for response fidelity in that local experiment. It is not a measured classification gain from inserting this second-order tensor into the continuous load/line bank, which uses switched local predictors instead.

The coefficients in (17) form a tensor over measured coordinate, elapsed time and source indices. The useful term is the physical cross-source interaction Q_{qnij}; the array's shape alone does not provide an accuracy or computational advantage. In the continuous estimator, M is instead a vector of aggregated source/family weights at each report, and ΔM is its temporal difference, as defined in (14)–(15). That implementation does not require a tensor decomposition or a tensor-learning library.

D. WHY INTEGRATE SEVERITY RATHER THAN SELECT THE BEST FIT?

An extra source with zero amplitude can reproduce a smaller source set. Best-fit residual alone therefore cannot decide cardinality. In the archived load-support estimator, the model-selection quantity is

$$Z_S = \int p(r \mid s_S(a), \Sigma) p(a \mid S)\, da, \qquad p(S \mid r) \propto Z_S p(S). \tag{18}$$

Integration accounts for the prior volume of amplitudes consistent with the data, rather than rewarding only the best-fitting amplitude. This is standard Bayesian model comparison applied to an electrical response model, not a new Bayes rule. The archived implementation evaluates the null, 16 single-source and 120 pair supports with 31-point Gauss–Hermite quadrature per amplitude. Table 3 shows both its successes and its loss of resolution for weak sources.

An audit found a one-source evidence normalization offset in this archive: the implemented cardinality prior is proportional to $(0.20, 0.50/(2\pi), 0.30)$ rather than the declared $(0.20, 0.50, 0.30)$. The table preserves the executed results under that effective prior; it does not validate calibrated

TABLE 3. Archived joint cardinality-and-support decisions. Onset, load family and operating context are supplied. These are noise replays of shared physical trajectories, not independent continuous detections.

Amplitude regime	Replays	Joint support Top-1
Finite / finite	1,440	100.00%
Moderate / moderate	1,440	98.82%
Weak / strong	1,440	66.25%
Weak / weak	5,760	24.38%

posterior probabilities or a corrected replay. The continuous estimator uses separate, independently checked local evidence code. Consequently these component results motivate the physical/Bayesian construction, but do not establish that adding Bayes or tensor storage alone causes its final accuracy advantage.

VI. EXPERIMENTAL DESIGN

The test asks whether one method can attribute six successive load or line interventions in a continuous trajectory, including an intervention that begins while the preceding response is still recovering. The plant is a fixed Julia/PowerDynamics IEEE 39-bus research candidate. Each parent is integrated once across all ON and OFF actions; differential states are not reset. PMU reports are exported at 30 frames/s, independently of solver steps. A switch-aligned report uses the post-switch algebraic value. RAW0001 informed the simulator retrospectively and is not an independent test record.

The sealed evaluation has 18 six-event continuous parents, 27 persistent line-opening parents and 12 normal parents. Every parent is replayed under matched R/X, an R/X change represented in the model bank and an R/X change outside it. Two instrumental-noise scales give 171 physical trajectories and 342 observed records. The 648 continuous event realizations repeat 108 interventions; the physical parent, rather than the PMU frame or replay, is the uncertainty unit. The earlier development examples in Fig. 8 are not part of this sealed comparison.

A. WHAT THE FEATURE BASELINE ACTUALLY DOES

The hackathon submission was a hierarchical detector, event classifier and typed localizer based on ExtraTrees. It summarized each 30-s PMU window with voltage/current magnitude and angle statistics, frequency, ROCOF, missingness, phase spread, approximate P/Q, rolling changes, RLS/Kalman level innovations and graph-temporal differences. Its selected bundle reports 45,162 input features and internal simulated-validation macro-F1 0.960 with exact localization 0.852. These numbers describe that archive's broader nine-label, window-level task; they are not scores on the present event-level cohort. The original submission is preserved.

To compare at 30 reports/s, an earlier development campaign built two *new*, frozen causal ExtraTrees adapters from 47 PMU coordinates. Each coordinate is centered and robustly scaled using the quiet prefix. For trailing windows of 1, 5 and 10 s, the reference adapter computes median, standard deviation, 10th/90th percentiles, a recent-quarter change and RMS first difference. This gives $47 \times 3 \times 6 = 846$ features per report. The short-window adapter adds the current value and mean, standard deviation and two lag changes at 3, 8 and 15 reports, giving 1,457 features in total. Both use 400 trees, balanced class weights, 19 classes (quiet plus 18 assets), and six-report confirmation; tree votes are scores, not calibrated event probabilities. Their training parents precede and are disjoint from the sealed cohort. Neither model was retrained or threshold-adjusted on these outcomes. The original split contains 32 local physical IDs for fitting (64 records, 78,784 training rows) and eight for validation (16 records). A later dependency audit maps these to 26 fitting and eight validation groups, with two normal-control groups shared across those sets. This limits independence of the internal validation; it is not evidence of leakage into the separate confirmatory cohort. The reference and short-window classifiers already obtain only 0/72 and 4/72 timely load attributions in that development validation. Their frozen performance therefore does not represent a successfully optimized load classifier.

A later grouped retraining study uses 31 fit, eight validation and 17 reserved physical groups: approximately 70/30 development/reserved, with an inner validation partition. That model is a separate artifact, not either comparator executed in the confirmatory study. Randomly splitting PMU reports 70/30 would allow correlated trajectories to cross partitions. RAW0001 is a known retrospective transfer check, not the reserved 30% or an external test. The supplement distinguishes these three evaluation contracts.

The sealed methods are the five-model physical bank, its nominal-network version and these two causal ExtraTrees adapters. All receive the same 30-frame/s PMU stream and retain their own frozen decision rules. This tests complete procedures at their declared operating points, not a matched-false-alarm or unlimited-supervision comparison. The physical banks also have different publication rules, so their contrast does not isolate network-model averaging.

Chen *et al.*'s published quickest-change detector [2] is eligible only for persistent line openings and normal records. The declared adapter uses nonoverlapping 15-Hz angle increments, a nominal single-line catalog and prefix-estimated drift and scale. It stops at its first alarm. Its DC and white-injection assumptions do not fully describe the simulator's colored PMU noise; no published average-run-length guarantee is transferred to these records.

For continuous streams, an alarm is matched temporally before inspecting its source. A timely decision requires the right family and asset before true OFF. Unmatched event-stream alarms are extras. Restoration messages on unmatched episodes are counted separately; this count is not every possible physical OFF-timing error. For persistent lines, the first stop is scored as correct, premature, wrong-source or censored. Failed records remain in the intended denominators. Paired 95% intervals resample physical parents 10,000 times. Logical delay uses report timestamps; wall-clock inference cost is a separate quantity. Full calendars, thresholds and

TABLE 4. Three distinct inference contracts. Historical hackathon validation and the sealed causal comparison have different units and cannot be subtracted as if they were the same score.

	Hackathon submission (historical)	Causal ExtraTrees comparators	Physical history bank
Input and clock	Eight PMUs; one decision per 30-s window	Same 47 PMU coordinates as physical bank; 30 reports/s	Same 47 coordinates; 30 reports/s
Information used	Engineered window features; supervised family/location labels	Trailing statistics; supervised active-asset labels from earlier simulated parents	Nominal DAE, candidate asset operators, PMU terminal map and quiet prefix
Decision	Hierarchical detector/classifier/localizer	Tree vote, then six-report episode confirmation	Conditional history evidence, active/cause masses and fixed publication rules
Tested scope here	Not rerun on the sealed cohort	Nine loads, nine lines, successive events	Same nine loads and nine lines, successive events
Main limitation	Window-level historical score is not a causal event-level score	May miss load sources absent from its learned feature boundaries	More extra alarms under out-of-bank R/X

reproduction details are in the supplement.

VII. DEVELOPMENT COMPARISONS AND WORKED SEQUENCE

A. DIRECT TRANSFER OF THE SUBMITTED HACKATHON SYSTEM

Before the confirmatory continuous study, the unchanged submitted ZIP and an earlier physical estimator were executed on the same 124 windows: 96 event realizations and 28 normal windows. Figure 6 preserves their actual family confusions. The submitted ExtraTrees pipeline detected 32 of the 96 events, assigned the right family to 23, and the right family and exact source to 0. The earlier physical estimator obtained 96/96 for all three quantities. Neither alarmed in the 28 normal windows. The source result was checked against the submission's output CSVs, not inferred from the family matrix.

This comparison exposes a transfer failure of the delivered system on the later simulator's load/line catalog. It does not show an official hackathon-score improvement. The submitted model retains its broader training domain and 30-s window rule, whereas the physical estimator knows the simulator and its 12 candidate origins. These were development cases, and the comparison permits both methods the same window endpoint. The next study therefore uses separately trained causal ExtraTrees comparators, successive events and a new frozen cohort. Its results must be read as a second experiment, not as more independent repetitions of the perfect development score.

B. FOLLOWING SUCCESSIVE EVENTS THROUGH RECOVERY

The example in Fig. 8 follows six physical interventions without restarting the network. With matched R/X, all six sources are identified before their OFF times. Under a network change outside the bank, an extra bus-20 diagnosis precedes the real bus-31 load; bus 31 is then diagnosed only after its physical restoration. The lower panel shows the mechanism visible to the method: active bus-31 mass rises late in the mismatched record. This is one development trajectory, not a performance estimate.

The development controls also show that deleting alarms is easy if true events may be lost (Fig. 7). The stricter onset witness has no extras but loses a real bus-16 intervention. The frozen five-model procedure was therefore evaluated with its original publication rule; no rule was revised from the sealed outcomes.

C. COMPARISON WITH A SIMPLER PHYSICAL PROCEDURE

An earlier development control uses the same nominal physical and PMU operators, one retained state and a short generalized likelihood-ratio search. With a common six-report publication clock, it obtains 37/54 timely decisions and 37 extras; the history mixture obtains 51/54 with 3. This is an executed physical comparator, separate from the later confirmatory cohort. Different search, severity treatment and score semantics prevent it from being a pure Bayes or memory ablation. Its equations and full table are included in the supplement.

VIII. SEALED EVALUATION: ACCURACY AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched event-stream alarms. Its nominal counterpart makes 643/648 with 92 extras. The frozen causal ExtraTrees reference makes 330/648 with 29; its short-window variant makes 352/648 with 16 extras. These are paired replays of 108 interventions, not 648 independent events. Figure 9 shows why the totals differ: the physical banks recover nearly all loads, while both learned adapters already recover most tested lines. The extra alarm burden of the physical banks rises sharply when true R/X changes fall outside their five-model bank.

A posthoc audit of the original episode matches further separates coverage from classification. Every matched diagnosis emitted before OFF has the correct family and source in all four procedures; their errors in the timely endpoint are missing or late diagnoses. The load advantage therefore

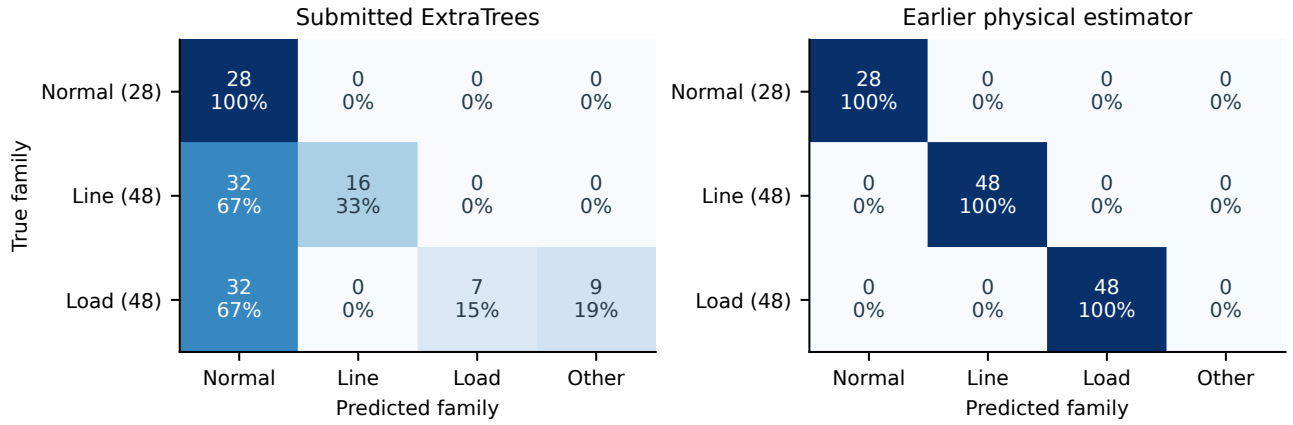


FIGURE 6. **Historical development transfer on shared observation windows. Cells show counts and row percentages. All out-of-scope predictions are retained as "Other"; the submitted system's nine such predictions were the unknown-event label. This is a family matrix, not a localization matrix. The physical comparator is an earlier estimator, not the final continuous bank. The 96 event realizations reuse 36 physical event trajectories; the normal windows come from six physical trajectories.**

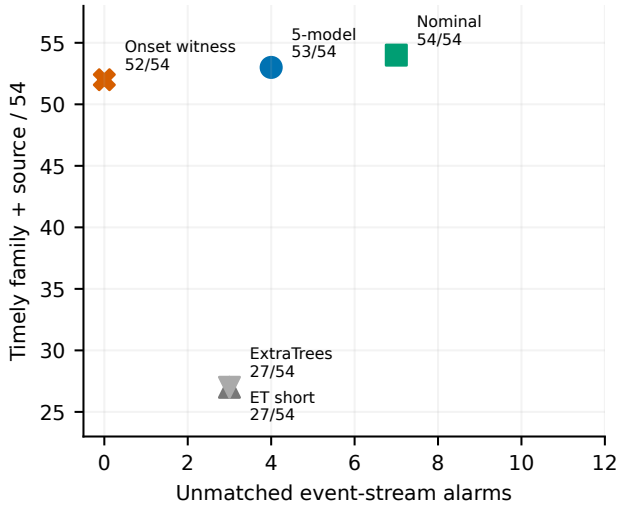


FIGURE 7. **Development decision trade-off on 18 interventions replayed under three R/X settings. A stricter onset witness eliminates extra alarms but misses a true bus-16 load. The 54 plotted realizations share physical parents. These controls were not selected using the later sealed evaluation.**

TABLE 5. **Sealed continuous-stream decisions. Each family has 324 replayed instances from 54 distinct interventions (108 across both families); extras are unmatched alarms on event records. The methods retain unequal alarm operating points.**

Method	Load	Line	Extras
Physical bank	322/324	322/324	72
Nominal bank	322/324	321/324	92
ExtraTrees	15/324	315/324	29
ExtraTrees short	30/324	322/324	16

concerns timely coverage. It does not demonstrate higher conditional classification accuracy among diagnoses already available before OFF. The supplement shows the complete confusion view and counts unmatched alarms separately.

The paired intervals in Fig. 10 confirm a timely-attribution advantage over this ExtraTrees reference in each tested stratum. They also show a higher rate of unmatched alarms under out-of-bank R/X. Those intervals resample physical parents, preserving correlation across noise replays and network conditions. They do not establish superiority at a common false-alarm operating point. The physical bank's stratum median diagnosis delay is about 0.27 s in matched and in-bank conditions; the ExtraTrees medians range from 0.19 to 0.30 s. There is no general latency advantage.

The separate persistent-line test scores the first stop. Both physical banks and both ExtraTrees adapters identify the right line first on 162/162 openings. The declared Chen2016 adapter does so on 74/162; 66 records stop before the opening and 22 stop afterward on a wrong line. A separate descriptive audit of the same persistent records shows 164 and 179 later unmatched alarms for the two ExtraTrees procedures, versus 0 for the physical bank. Correct first identification therefore does not imply stable maintenance of a sustained event. This audit does not change the first-stop comparison or apply a recurrent score to Chen. Chen also stops on all 72 paired normal records, whereas the continuous methods alarm on none. Chen's normal exposure ends at its first stop (2,522 s); the other methods monitor 42,336 s each. These unequal exposures and the adapter's white-injection assumption prevent a general claim against published quickest-change detection. Full stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

The experiment establishes performance on the specified nine-load, nine-line simulator catalog. It does not establish a new hackathon score, unseen-source generalization, calibrated event probabilities, field operation or real-time throughput.

IX. CONTROLLED COMPONENT STUDY

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank,

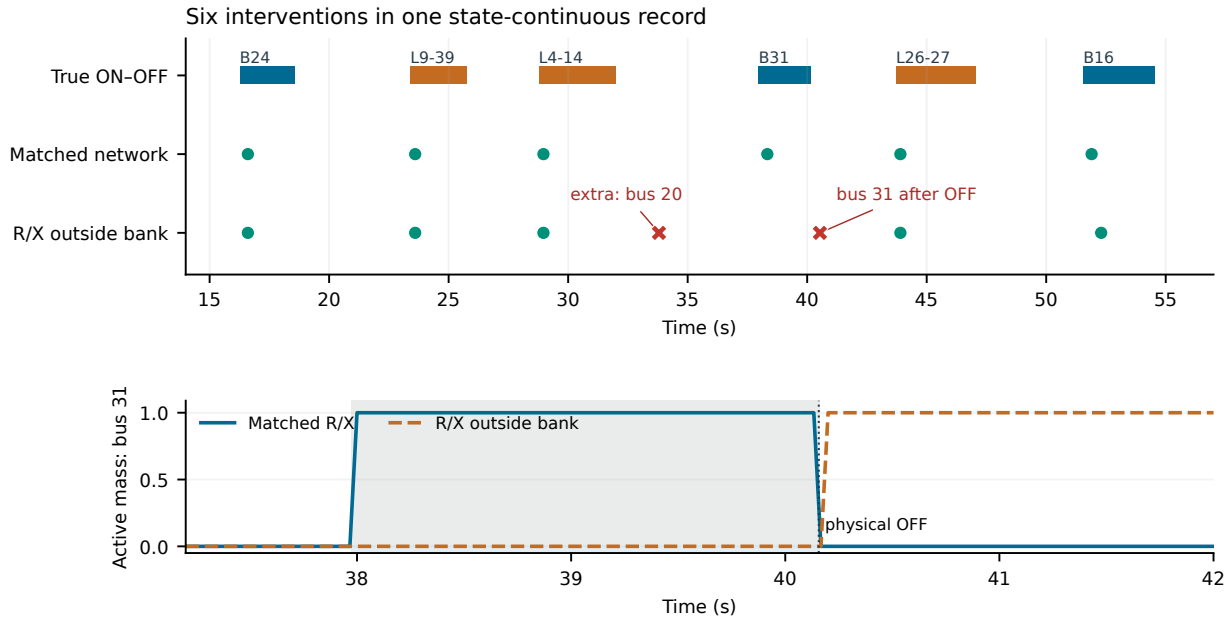


FIGURE 8. **One development sequence. Bars show reserved physical ON–OFF truth; circles are correct diagnoses emitted before OFF; crosses mark an extra bus-20 alarm and a bus-31 diagnosis emitted after OFF. Both lower decision rows use the same physical bank under different R/X conditions. The lower panel shows approximate active mass for bus 31: with R/X outside the bank, the correct explanation becomes dominant only after restoration. Truth is shown retrospectively and never enters inference.**

we fixed a subsequent 2×2 study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{19}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{20}$$

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k / v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

A. EXECUTION STATUS

The component study is in progress. No component-effect result is claimed in this draft. The original continuous-test outcomes remain unchanged.

X. INTERPRETATION AND LIMITS

A. PREDICTION IS NOT IDENTIFICATION

For the exact observation map Ψ, compare a new intervention e with and without the same earlier history h:

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{21}$$

Both paths start from the same physical and instrumental state. For an additive intervention under the same linear time-invariant dynamics, this increment is independent of the prior state. A topology change can alter the state matrix, so that statement does not apply to switched linear models in general. Neither switching nor nonlinearity guarantees more source information. Our stream implementation approximates both

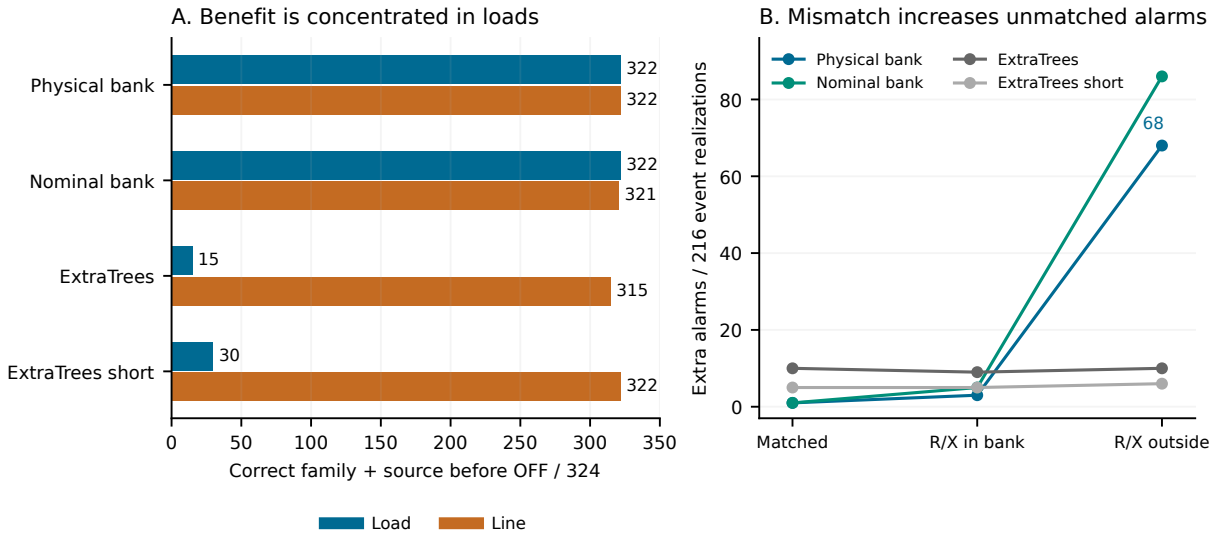


FIGURE 9. **Sealed continuous evaluation. (A) Timely correct family and source by event family, with 324 paired realizations per family. (B) Unmatched event-stream alarms across the two noise levels within each R/X condition, with 216 event realizations per condition. The plots display the gain in load attribution and the alarm cost under out-of-bank network error together. Each method retains its own frozen publication rule and false-alarm budget.**

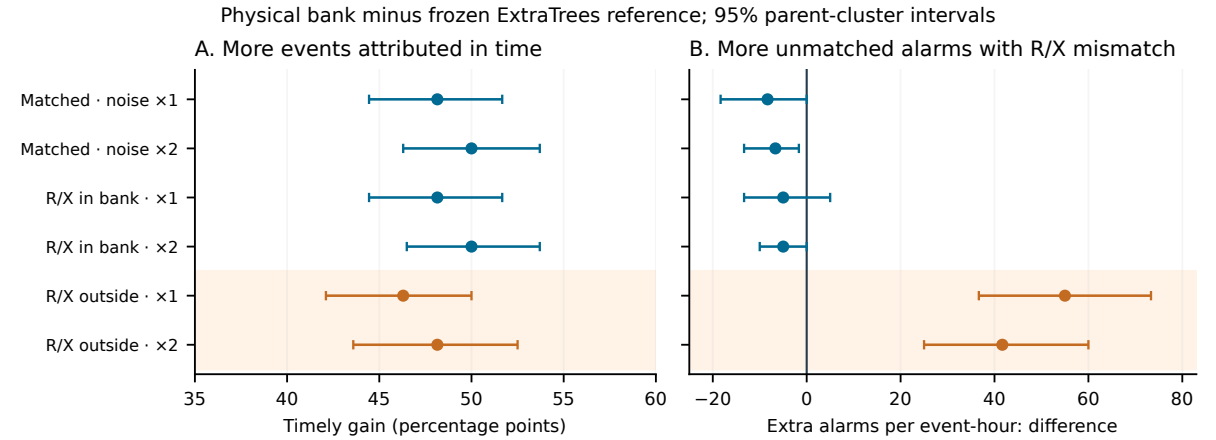


FIGURE 10. **Physical bank minus frozen ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per event-record hour, whose denominator includes the 12-s startup. The orange rows expose the out-of-bank R/X failure. The intervals describe this finite simulated population.**

paths with local switched predictors. It does not solve the full nonlinear DAE for every history.

A good fit to one source does not establish that competing sources can be separated. With severity and nuisance coordinates β_j profiled over their admitted set $\mathcal{A}_j$, a local separation measure is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{22}$$

Small distance makes location noise-sensitive; it is neither a universal error bound nor a calibrated posterior [9], [10]. Fit, source ranking and uncertainty calibration require separate tests. The supplementary controls retain their known-onset and observation contracts; their state reconstruction scores are not pooled with the causal study. Severity fitting can absorb context error parallel to a source signature while leaving intensity biased. This standard projection effect explains why correct location and inaccurate response amplitude can coexist.

B. WHAT RECOVERY MEMORY ESTABLISHES

A separate full-DAE control supplies the earlier history and later onset. Source separation stays essentially unchanged when that history is known and falls when prior severity is uncertain; binary source accuracy has no mean gain. Retaining recovery is therefore justified as a consistent prediction of the continuing response, not as a universal increase in location information. The stream comparison tests the complete frozen procedure. Its admission-rule ablation does not isolate every operation that retains recovery.

A separate signature test holds pre-event measurements and offset equal while changing the event–context response

term. It isolates a response approximation, not an improvement of the unchanged causal detector. Known timing, one scalar context and four load sources bound that result.

C. WHAT THE COMPARISON DOES NOT ESTABLISH

Trajectory libraries, sequential likelihoods and multiple-model filters precede this work. The claim here is the specified competition of new interventions with retained recovery, and its measured behavior, rather than a general filtering optimum. Warner's trajectory procedure was not run on this cohort. The feature baseline's trailing summaries describe how signals changed, but do not encode the counterfactual response of *each candidate source* through the network and PMU terminal map. That is a concrete difference in information used by the algorithms and is consistent with the observed load gap; these experiments do not prove that a better feature set or larger training set could not close it. The original hackathon submission addressed nine labels with 30-s decisions. It is a useful earlier result and a source of feature design, but its internal validation score cannot be compared numerically with event-level timely attribution here. The two causal ExtraTrees adapters were trained separately for this task; their load misses do not establish a new official hackathon score or rule out better supervised models. Chen's first-stop adapter is tested only on persistent lines under DC and white-injection assumptions.

D. BROADER HACKATHON SCOPE AND RETROSPECTIVE TRANSFER

The earlier all-family pipeline and integrity interface are retained as separate results. An archived observation-only composition reproduces all 180 physical/integrity windows and reconstructs availability over 161379 reports; this verifies interface consistency, not additional detection accuracy. On retrospectively annotated RAW0001 windows, a source-only learned hybrid changes joint family/source correctness from 1/24 to 2/24, with one gain and no losses. It is not the physical bank and does not establish broad transfer. The frozen physical bank rejects that record because the prefix ROCOF noise scale is outside its admitted support; no later quiet prefix was substituted to force admission. The supplement preserves these outcomes alongside the simulation results.

E. SIMULATION AND DEPLOYMENT BOUNDARY

The causal catalog has nine load buses and nine lines, with a quiet startup. Missing PMUs, unknown families, unrestricted simultaneous sources, unseen-source transfer and field records remain untested. The generating plant and inference predictors share the F01 equation family and PMU map. The three R/X conditions perturb parameters within that family; they do not test a different producer, control law or instrument installation. Thus the high within-simulator hit rate is not evidence of the same rate on RAW0001 or a field network. The simulator keeps a provisional BUS5 current terminal and omits BUS39 current angle. Its noise includes temporal and cross-PMU dependence within signal families but omits the archived cross-family factor; frequency/ROCOF noise is added after phase-based estimation as separate signal families. It is not obtained by differentiating the exported noisy angle, and the archived cross-family noise factor is not applied. Hence agreement with this simulator does not verify the source producer's CT placement or a complete physical PMU installation.

Out-of-bank R/X preserves most timely source decisions but sharply raises unmatched alarms. The parent-cluster intervals describe these simulated parents, not all grids or PMU layouts. Context fitting, history pruning and shared covariance approximations also prevent a calibrated-posterior claim. Report-causal operation has not been timed as a real-time service; the implementation is not validated for protection or control.

XI. CONCLUSION

The proposed procedure attributes load and line interventions by comparing their predicted PMU responses with the observed stream. Each candidate starts from an estimated past state. Restoration changes its input or topology while its physical state continues to evolve. The method integrates local load severity and reports event activity separately from the cause of a continuing response.

The sealed continuous evaluation gives 644/648 timely family/source decisions with 72 extra alarms for the mixture, and 643/648 with 92 for nominal. The frozen ExtraTrees reference gives 330/648 with 29 extras. Most mixture extras arise under out-of-bank R/X, so the gain in timely attribution does not imply uniformly better alarm control.

The component experiments preserve useful earlier findings: network conditioning recovers hidden voltage information, graph-temporal features improve a historical learned localizer, and mixed-source terms reduce local response error. They do not establish an isolated causal gain from Bayes or tensor storage in the final estimator.

The earlier hackathon pipeline remains a broader nine-label, feature-based system. Its 30-s-window validation is a separate result; the causal ExtraTrees successor in this paper provides the direct same-stream comparison. The physical bank's strongest supported gain is timely load-source attribution, while learned features already diagnose most lines in this catalog with fewer unmatched alarms.

The evidence concerns a specified load/line catalog in one simulator. Unmodeled R/X increases the alarm burden. The controlled component study addresses amplitude scoring and state carry at inferred closure; it does not remove all Bayesian estimation or all temporal memory. All-family diagnosis, calibrated event probabilities, field transfer and real-time deployment remain outside the demonstrated scope.

A source-excluded comparison would test whether the procedure generalizes beyond its supported catalog. A learned correction of physical residuals could address model mismatch, provided it is evaluated independently of source-label fitting. Joint state/context uncertainty and event-probability calibration also require further validation.

REFERENCES

[1] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.

[2] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.

[3] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.

[4] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023, posted June 16, 2023; cited as a preprint.

[5] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.

[6] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.

[7] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.

[8] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.

[9] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.

[10] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.

[11] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.

[12] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 6, pp. 1140–1153, 2014.

[13] R. Anguluri, O. Kosut, and L. Sankar, "Localization and estimation of unknown forced inputs: A group LASSO approach," *IEEE Transactions on Control of Network Systems*, vol. 10, no. 4, pp. 1997–2009, 2023.

[14] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.

[15] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.

[16] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.

[17] H. A. P. Blom and Y. Bar-Shalom, "The interacting multiple model algorithm for systems with markovian switching coefficients," *IEEE Transactions on Automatic Control*, vol. 33, no. 8, pp. 780–783, 1988.

[18] Y. Boers and H. Driessen, "A multiple model multiple hypothesis filter for Markovian switching systems," *Automatica*, vol. 41, no. 4, pp. 709–716, 2005.

[19] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.

[20] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.

[21] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.

[22] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.

[23] M. Ensaf and M. Barati, "Harnessing kernel regression for stochastic state estimation in solar-integrated power grids," arXiv preprint arXiv:2309.10200, 2023, version 1. [Online]. Available: https://arxiv.org/abs/2309.10200

[24] A. M. Tartakovsky, T. Ma, D. A. Barajas-Solano, and R. Tipireddy, "Physics-informed gaussian process regression for states estimation and forecasting in power grids," *International Journal of Forecasting*, vol. 39, no. 2, pp. 967–980, 2023.

[25] A. Alalem Albustami, A. F. Taha, and S. Mahadevan, "DAE-aware bayesian inference for joint generator-network parameter estimation," arXiv preprint arXiv:2604.15686, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2604.15686

[26] R. Nagi, X. Huan, and Y. C. Chen, "Bayesian inference of parameters in power system dynamic models using trajectory sensitivities," *IEEE Transactions on Power Systems*, vol. 37, no. 2, pp. 1253–1263, 2022.

JORGE LUIS MAYORGA TABORDA studied electronic engineering at Universidad de los Andes, Bogotá, Colombia. His academic work has addressed distributed control using population dynamics, including applications to urban traffic systems.

YESSICA AFRICANO RODRIGUEZ carried out undergraduate thesis work on photovoltaic integration at Universidad de los Andes, Bogotá, Colombia. Her published research includes co-simulation and stochastic assessment of photovoltaic hosting capacity in distribution systems.



MASOUD BARATI received the Ph.D. degree in electrical engineering from the Illinois Institute of Technology in 2013. He is with the Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA. His research includes mathematical modeling and optimization of electric power systems.



PAYMAN DEHGHANIAN received the Ph.D. degree from Texas A&M University in 2017. He is an Associate Professor with the Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA. His research includes power-system reliability, resilience, and data-informed decision-making.



DAVID FELIPE CELEITA RODRÍGUEZ received the Ph.D. degree in engineering from Universidad de los Andes, Bogotá, Colombia. He is an Associate Professor and Director of Outreach and Engagement with the Faculty of Engineering, Universidad de La Sabana, Chía, Colombia. His research includes power-system protection, fault location, and real-time simulation.

Digital Object Identifier

Supplementary Material for Physics-Informed Bayesian Event Attribution From Sparse PMUs

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]

[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

Corresponding author: Jorge Luis Mayorga Taborda (e-mail: jl.mayorga236@uniandes.edu.co).

ABSTRACT This supplement gives the executable statistical contract of the causal physical-history method: local DAE reduction, candidate transitions, truncated-severity evidence, state updates, restoration, worked examples, and scenario-level reproduction. It also defines the measurement-integrity interface and separates causal alarm, diagnosis and restoration metrics from the historical hackathon window evaluation. Earlier reconstruction, batch-source and proposed joint-estimator studies are preserved in the separately compiled Historical Technical Notes. Their results do not enter the causal method's denominators.

INDEX TERMS phasor measurement units, physical event attribution, measurement integrity, causal inference, reproducibility.

APPENDIX A HOW TO READ AND REPRODUCE THE CURRENT METHOD

This companion begins with the recovery-aware causal method used in the current article. Sections A–C give its execution order, local derivations, worked examples and simulation contracts. The separately compiled Historical Technical Notes preserve full diagnosability proofs, earlier state-reconstruction and batch-estimator studies, and secondary evidence. The earlier studies support particular components; they are not additional tests of the complete causal algorithm.

The article retains the physical model, the defining evidence integral, the recovery mechanism, the decision outputs and decisive results. This supplement supplies the intermediate calculations. The Historical Technical Notes develop functional observability, nuisance-profiled information and source ambiguity, and distinguish the implemented batch estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint estimator $\mathcal{E}_{\mathrm{J}}$. Neither should be silently substituted for the causal history bank, archived as R11. The conditional Gaussian updates use standard filtering identities [1]; the derivation below specifies their role and the approximations made by the implemented bank.

A. INPUTS, OUTPUTS AND THE INFORMATION BOUNDARY

The inputs are successive reports from eight PMUs, the fixed physical and instrumental operators, the admitted load/line catalog and the frozen method configuration. The output is a timestamped sequence of detection, family/source and restoration messages, together with retained attribution masses. True ON/OFF times, true sources and generating severities belong to the evaluator. They are not inputs to the causal algorithm.

Three times must remain distinct: the latest observed report, the estimated physical onset, and the public emission time. Looking back to propose an onset is compatible with causality when the message is emitted only after the required reports arrive. It does not permit backdating the public alarm.

B. EXECUTION ORDER FOR ONE STREAM

1) **Load the frozen operators.** Retain voltage reference, oriented terminal currents, load input directions, switch modes and instrument memory. A line opening changes the network and observation operators. It is not represented by fitting a load amplitude.
2) **Fit the startup context.** Use the first 300 reports for operating factors, common reference and noise scale. For each network hypothesis, set the voltage/current discrepancy to the difference between the observed prefix mean and its reference prediction. Keep that offset fixed afterwards. Frequency and ROCOF use their empirical prefix centers. The prefix is not scored again as independent model-selection evidence.
3) **Predict the next report.** Propagate the retained histories and evaluate each network model's predictive den-

sity before correcting states or choosing new histories with the current report.

4) **Propose candidate onsets.** The frozen screen uses lags 0–18 reports within a ten-second stored state history. Each proposal starts from a stored state before its proposed onset, including any recovery already present. The bank does not enumerate every possible onset in the stored interval.
5) **Score histories.** For loads, integrate severity with the bounded prior and the local conditional Gaussian likelihood. For lines, score the admitted switched operator. Score a report before using that same report for its conditional state correction.
6) **Update and compress.** Update severity and conditional states, network weights, restoration children and the retained beam. Record discarded mass. At episode closure retain the augmented-state mixture mean; the omitted between-history spread is a documented approximation, not an exact posterior covariance.
7) **Publish and record.** Aggregate active mass M, cause mass Γ and ΔM. Apply the frozen detection, diagnosis and restoration rules. Save emission time and the last report used, in addition to estimated onset and source. Continue propagation through OFF.

C. WHY REPLAYED REPORTS ARE NOT NEW EVIDENCE

Let a retained parent have log score $\ell_{0:k} = \ell_{0:\tau-1} + \ell^{old}_{\tau:k}$. Replacing its suffix by a newly proposed physical history gives

$$\ell^{new}_{0:k} = \ell_{0:k} - \ell^{old}_{\tau:k} + \ell^{new}_{\tau:k}. \tag{1}$$

Transition and prior terms must accompany their corresponding history. The rolling screen supplies a proposal, not an extra independent likelihood. Adding the new suffix without removing the old suffix would count the same reports twice. This bookkeeping statement does not make adaptive screening or beam pruning exact Bayesian inference.

D. FROZEN NUMERICAL CHOICES

The input grid is 30 reports/s. The five network hypotheses are nominal, global $\pm 2\%$ and selected-branch $\pm 3\%$ paired R/X perturbations. Each hypothesis is fixed in time; the weights change. They do not identify a continuous impedance posterior.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$. Birth mass is divided equally between load and line families, then uniformly among the admitted sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over intervals bounded by $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can create OFF children at either half-report position, inheriting state and severity. After likelihood normalization, the beam retains the union of the 96 highest-weight recent histories, 64 older histories and at least one per occupied source. The union is renormalized.

The internal observer confirms active-mass evidence at 0.99 for six reports and has a separate 15-report internal OFF clock. The public R11 publisher requires an internal pending episode and six reports of source-qualified active support above 0.99; its public OFF requires six reports of the diagnosed cell's joint cause-and-restoration mass above 0.99. The detailed source qualification, diagnosis and replacement rules are given below. These thresholds are implementation settings, not guarantees of probability calibration or false-alarm control. Nominal and mixture use their frozen publication paths, so their comparison does not isolate network averaging alone. The admission-rule ablation changes publication, not the recovery dynamics.

E. SOURCE QUALIFICATION AND PUBLIC EPISODE RULES

The source screen and the onset search are distinct. Source qualification uses a 30-report (one-second) conditional innovation vector v with 47 coordinates per report. Fixed normal-mode gains propagate the stored pre-window state and whiten the suffix. For each prepared response s_j and lag from 0 to 25 reports, let

$$\begin{aligned} g_j &= s_j^\top s_j, \qquad u_j = s_j^\top v, \qquad R = v^\top v, \\ \widehat{a}_j &= \mathrm{clip}(u_j / \max(g_j, 10^{-30}), -0.2, 0.2), \\ G_j &= 2\widehat{a}_j u_j - \widehat{a}_j^2 g_j, \\ F_j &= \frac{\max(G_j, 0)}{\max\{(R - G_j)/1409, 1\}}. \end{aligned} \tag{2}$$

Line columns use fixed amplitude one. The source score is the maximum over its columns; the threshold is the $1 - 0.001/936$ quantile of an $F_{1,1409}$ distribution, approximately 24.0115. This threshold and the variance floor are a frozen proposal/publication rule, not a calibrated global false-alarm probability. The statistic is never multiplied into the Bayesian history likelihood. This one-second qualification screen does not enlarge the separate birth search beyond its 0–18 report lags.

Define $b_c = 1$ for a line, and $b_c = \mathbf{1}\{F_c > 24.0115\}$ for a load. At each monitored report, the publisher updates consecutive-report counters for $M_c b_c \geq 0.99$, $\Gamma_c \geq 0.99$, and $\Gamma_c P(\mathrm{OFF} \mid c) \geq 0.99$. A false condition clears its counter. An internal alarm opens a pending episode. A public alarm then requires either six pending reports with $\sum_c M_c b_c \geq 0.99$ or an already accumulated six-report line-active counter. The latter permits line evidence collected before the internal pending message to count.

A public episode receives one diagnosis when its highest cause-mass cell has six consecutive reports above 0.99, or at the 15-report deadline after the public alarm. The output family maximizes summed cause mass; sources are ranked within that family. The diagnosed source becomes the episode owner. Six reports of that owner's joint cause-and-OFF mass above 0.99 emit a public restoration. These publication operations do not rescale M or reset the physical state.

The internal mixture lifecycle separately replaces an episode after six reports with mass at least 0.99 on active

histories whose onset is later than its alarm (using the implementation's quarter-report timing guard). It withdraws an unconfirmed episode after the diagnosis delay when null mass has exceeded 0.99 for 15 reports. Its internal restoration requires cause mass at least 0.99 and conditional retired mass at least 0.99 for 15 reports. These terminal messages are propagated to the public episode if already emitted; an unconfirmed pending episode is recorded as rejected. Finalization marks any remaining public episode right-censored. The nominal comparator retains its own fixed internal lifecycle, so the nominal/mixture comparison is not an isolated model-averaging ablation.

APPENDIX B LOCAL CALCULATIONS AND WORKED EXAMPLES

A. COMPLETING THE BOUNDED-AMPLITUDE INTEGRAL

For one retained severity interval, write $e = sa + \epsilon$, with $\epsilon \sim \mathcal{N}(0, S)$, $S \succ 0$, and $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, $v > 0$. Here μ, v are parameters of the underlying untruncated Gaussian, not the bounded distribution's moments. With $g = s^\top S^{-1} s$ and $q = s^\top S^{-1} e$, expansion gives

$$\begin{aligned}(e - sa)^\top S^{-1}(e - sa) + (a - \mu)^2/v \\ = e^\top S^{-1} e + \mu^2/v - \mu'^2/v' + (a - \mu')^2/v', \quad (3)\\ v' = (v^{-1} + g)^{-1}, \qquad \mu' = v'(\mu/v + q). \quad (4)\end{aligned}$$

Define $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Integrating the last Gaussian square over $[l, u]$ yields

$$\begin{aligned}\log L = & -\tfrac{1}{2}[e^\top S^{-1} e + \log|S| + n_y \log(2\pi)] \\ & + \tfrac{1}{2}[\log(v'/v) + \mu'^2/v' - \mu^2/v] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \quad (5)\end{aligned}$$

The posterior on that interval is $\mathrm{TN}_{[l,u]}(\mu', v')$. With $\alpha = (l - \mu')/\sqrt{v'}$, $\beta = (u - \mu')/\sqrt{v'}$, $Z' = \Phi(\beta) - \Phi(\alpha)$ and standard normal density ϕ, its mean is

$$\mathbb{E}[a \mid e] = \mu' + \sqrt{v'}\,\frac{\phi(\alpha) - \phi(\beta)}{Z'}. \quad (6)$$

Thus neither the untruncated mean μ' nor a best-fit severity can simply replace the integral. The implementation combines retained intervals using their weights. The identity is exact for this conditional Gaussian model; it does not remove linearization, shared-covariance or pruning error.

B. NUMERICAL EXAMPLE OF ONE INTERVAL

This dimensionless example illustrates the calculation; it is not an IEEE-39 simulation or a new performance result. Set $S = 1$, $s = 2$, $e = 1$, $\mu = 0$, $v = 1$ and $[l, u] = [0, 1]$. Then $g = 4$, $q = 2$, $v' = 0.2$ and $\mu' = 0.4$. Substitution in (5) gives $L \simeq 0.342688$ and $\mathbb{E}[a \mid e] \simeq 0.464941$. The different posterior mean follows from truncation. The script scripts/verify_supplement_examples.py checks the closed form against direct numerical integration, including the posterior mean. It checks algebra only and does not modify a frozen experiment.

C. CONDITIONAL STATE CORRECTION

Represent the conditional state mean as $m_h + d_h a$. Let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the archived one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With shared covariance P_ρ in discrepancy regime ρ,

$$\begin{aligned}S_o = R + J_o P_\rho J_o^\top, \qquad & K_o = T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ = T_o m_h + g_o + K_o e, \quad & d_h^+ = T_o d_h + u_o - K_o s. \quad (7)\end{aligned}$$

To see the last line, apply the correction to a fixed amplitude:

$$T_o(m_h + d_h a) + g_o + u_o a + K_o(e - sa) = m_h^+ + d_h^+ a.$$

The severity distribution is updated by the same likelihood integral. Regime covariances come from nominal-operator Riccati equations. R11 does not propagate an independent exact covariance for every switched history. At closure the mean is retained and the omitted mixture spread recorded.

D. RESTORATION IS NOT A STATE RESET

Consider the illustrative scalar system $\dot{x} = -x + u$, $y = x$, with $x(0) = 0$. Apply $u = a$ for $0 \le t < \kappa$ and restore $u = 0$ afterwards. Direct integration gives

$$x(t) = \begin{cases} a(1 - e^{-t}), & 0 \le t < \kappa, \\ a(1 - e^{-\kappa}) e^{-(t-\kappa)}, & t \ge \kappa. \end{cases} \quad (8)$$

Both expressions agree at κ. For $a = 1$ and $\kappa = 2$, $x(2) = 0.864665$ and $x(3) = 0.318092$ even though the intervention is OFF. Resetting the reference to equilibrium would discard that predictable recovery. This example explains state continuity; it does not claim that one exponential describes the multimode IEEE-39 response.

E. ACTIVE MASS, CAUSE MASS AND MODEL REWEIGHTING

For one model, suppose an illustrative cell has mass 0.60 in active histories and 0.30 in restored histories, with 0.10 elsewhere. Its active mass is $M_c = 0.60$, while its originating-cause mass is $\Gamma_c = 0.90$. Recovery can therefore remain attributable without declaring the input active. These assigned masses illustrate definitions, not calibrated probabilities from an experiment.

For two reports, adding and subtracting $\sum_\ell w_{\ell,k} M_{\ell,c,k-1}$ proves

$$\begin{aligned}\Delta M_c = & \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ & + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \quad (9)\end{aligned}$$

For example, unchanged conditional masses $(0.8, 0.2)$ and network weights changing from $(0.5, 0.5)$ to $(0.75, 0.25)$ move M_c from 0.50 to 0.65. The entire 0.15 change comes from model reweighting. A rising aggregate mass need not mean that each model found stronger within-model evidence.

APPENDIX C SIMULATION SCENARIOS AND REPRODUCTION ORDER

A. PLANT, MEASUREMENT AND PROVENANCE

The active simulator is the frozen research candidate `SIM0001_JuliaPD_F01`, not an officially validated equivalent of RAW0001. Its contract fixes Julia 1.11.9, PowerDynamics 5.0.0, NetworkDynamics 1.3.0 and IEEE39 0.1.0. Initial states, effective parameter vectors, terminal map, reporting operator and stochastic model come from that snapshot. Nominal inventory files do not replace the effective operating point. The load law is the frozen impedance law.

The voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. Each current is an oriented branch-terminal measurement, not a nodal injection. BUS5 retains the provisional historical 5–6 branch, terminal 5, hypothesis; producer CT identity is unconfirmed and BUS39 remains partially validated. Execution uses full-precision configuration files rather than rounded tables. RAW0001 provided retrospective calibration context. RAW0002 remains outside these studies.

Each physical parent is integrated continuously through its interventions. Differential states remain continuous at switching, while algebraic values are recomputed consistently. Reports use $t = k/30$ with endpoints included; the switch-aligned report uses the post-switch algebraic value. Solver steps are recorded separately and observations are not interpolated across a switch. The causal instrumental history is preserved through OFF.

B. CONTINUOUS DEVELOPMENT AND CONFIRMATION

H03 and H04 each contain three 80-s event parents with six interventions and one 80-s normal parent. Crossing the four parents with three R/X conditions gives 12 physical trajectories, 18 unique interventions and 54 event realizations per campaign. Both are development evidence in the current article. Repeated conditions are not independent event draws.

C01 prespecifies 18 continuous parents, 27 persistent-line parents (nine lines, three repetitions), and 12 normal parents. The three R/X conditions produce $3(18 + 27 + 12) = 171$ intended physical trajectories; two instrumental-noise scales yield 342 observed records. The continuous parents contain $18 \times 6 = 108$ unique interventions. Their six condition/noise realizations are repeated observations of those parents. The 120-s and 600-s grids contain 3601 and 18001 reports, respectively. Completed results appear in the dedicated confirmation tables.

Continuous evaluation pairs an alarm temporally before inspecting its predicted label. The first unmatched post-ON alarm is eligible up to the next ON, with the inherited 30-s cap. Timely attribution requires correct family and asset emitted before physical OFF. Unmatched alarms, false restorations and open episodes are retained. Persistent-line evaluation instead scores the first stopping decision, including early stops, wrong sources, omissions and censoring. Chen2016 is eligible on that lane; it has no load or restoration output. Confidence intervals use 10,000 paired parent-cluster resamples, not independent PMU samples.

C. SEPARATE SIGNATURE AND RECOVERY CONTROLS

The E_i/X02 signature study uses a single load at buses $\{3, 7, 16, 24\}$, a 10-s record, known ON at 3 s and OFF at 6 s, and one scalar operating-context direction. It contains 57 dictionary/check trajectories and 85 test trajectories (80 interventions and five controls), with three common noise seeds producing 255 observed test records. The context and severity are estimated from measurements, while onset and family are supplied. The generating source is used to assess the response approximation and severity error. Those conditional metrics do not measure unknown-onset causal attribution.

The X04 recovery geometry control uses 38 trajectories. Prior load histories at buses 7 and 24, followed by delays of 0.5 and 2 s, give four recovery contexts. Three source-pair comparisons within each context give 12 correlated comparisons. The prior input is active from 2 to 3 s; the subsequent response is examined over 1.5 s. The supplied history, onset, family and two-candidate universe define an oracle control, with one uncertain prior-severity direction. It does not evaluate R11, unknown onset, event cardinality or operational false alarms.

D. REPRODUCTION WITHOUT CHANGING THE EXPERIMENT

1) Read each campaign's frozen plan and manifest. Select the intended population before interpreting any denominator. Keep current causal, signature, recovery-control and historical batch studies separate.
2) Resolve the F01 state, parameter and measurement files from its snapshot. For any new simulation create a new run directory; a new hypothesis requires a successor candidate. Do not run historical preparation scripts inside a frozen baseline.
3) Preserve ON/OFF continuity, report endpoints, noise seeds and the boundary between observations and reserved event truth. Run inference before releasing truth to the evaluator.
4) Retain solver/inference failures in intended denominators. Aggregate at the declared parent level and preserve the original comparators and publication rules. Do not retune from confirmation results.
5) Rebuild the two documents from the versioned tables and figures with scripts/build_manuscript_bundle.ps1. Its default compilation does not regenerate scientific artifacts or import new campaign results. An explicit artifact refresh is a separate operation.

The repository keeps the current walkthrough in this appendix, the frozen campaign evidence under `generated/frozen/`, and the earlier mathematical development in the separately compiled Historical Technical Notes. A successful PDF build verifies document consistency; it is not a new scientific validation.

APPENDIX D MEASUREMENT INTEGRITY AND THE HISTORICAL SUBMISSION INTERFACE

An electrical intervention and a measurement failure are separate objects. The physical output identifies a family and asset; the integrity output identifies missing or corrupt PMU observations. Missingness alone does not establish malicious intent. A concurrent physical event and missing PMU can therefore have two different locations.

The successor interface reuses the fitted detector, classifier and localizer from the actual submitted archive, rather than a similarly named exploratory pipeline. Archive member hashes are checked before inference. The input schema admits timestamps, three-phase voltage/current magnitudes and angles, frequency, ROCOF and the observed availability flag. Event labels, locations and annotated ON/OFF times are excluded. The fitted feature lists are also checked for label fields. This distinction matters because a historical diagnostic utility estimated a baseline from labeled normal samples over the complete record; that utility is not part of this inference interface.

The submitted classifier uses a physical-family head, a bad-data head and a missing/composition head with fixed rules. Its E5 and E7 outputs refer to missing and bad-data PMUs, respectively. E6 denotes missing data together with a physical prediction; its legacy location is the physical source, not the missing PMU. In the archived configuration, both composition and physical-confidence thresholds are zero, so E6 requires a nonzero physical prediction in a window with missing data; it is not a calibrated test of concurrent causation. E8 is the legacy mixed-physical proxy class, not an open-set rejection guarantee.

The availability interface evaluates each received report without looking ahead. It retains the affected PMU and channel mask rather than replacing missing measurements with future samples. The historical bad-data classifier uses a complete window and reports its result at the end. Within-window interpolation and centered features are permitted only under that delayed decision contract. Broadcasting the result to earlier timestamps does not make those earlier labels causal. The inherited cyber/integrity component is a comparator and integration component, not a newly claimed detector.

The implemented composition has two temporal scopes. Received-report availability messages update the current missing-PMU state. Completed-window records describe only their observed intervals; they retain separate physical and integrity fields and cannot assert current corruption or physical restoration. Their joint emission is the later component emission, with each original time retained. For example, a window that contains missing data can finish after the PMU has resumed reporting; publishing that interval must not reopen the missing episode. Likewise, an earlier ML source diagnosis does not advance the submitted classifier's window-completion time.

The sealed stream contains 206 messages, including 180 completed windows. Post-seal projection checks reproduce the physical source, integrity diagnosis, source-selection provenance and unavailable states of the components, and recover current availability on all 161379 reports without disagreement. The comparison retains the original component evaluations. It neither reruns inference nor creates a combined accuracy from report-level missingness and window-level physical diagnosis. The frozen physical filter remains explicitly unavailable.

APPENDIX E ALARM, DIAGNOSIS AND RESTORATION ARE DIFFERENT DECISIONS

Let τ and ω be true ON and OFF, and let t_A, t_D and t_R denote the emission times of an alarm, source diagnosis and restoration. The detector can signal before it has a source. An eventual correct source at $t_D > t_A$ is not a correct location already available at t_A. Timely attribution requires a correct family and source at $t_D < \omega$. Estimated onset times never replace emission times in a delay metric.

An event is associated with the first eligible unmatched alarm using time, before inspecting the predicted label. Incorrect labels remain associated errors rather than being reassigned to another event to improve the score. An alarm without a diagnosis, a wrong diagnosis, and no alarm are distinct outcomes. Restricted delay includes failures up to the evaluation horizon; conditional delay among successes is reported separately.

A restoration of an unmatched episode and a premature restoration of a matched event are also different errors. With a tolerance δ fixed before scoring, premature restoration is $t_R < \omega - \delta$; its lost active duration is $[\omega - t_R]_+$. Signed restoration delay is $t_R - \omega$. Report its distribution, missing closures and episodes still open at the horizon; a median alone can hide early closures.

The original C01 evaluator is preserved. Its false-restoration counter counts restorations of unmatched episodes, and its first-correct field can use the diagnosis associated with the first alarm even when emitted later. The successor evaluation reports these quantities with their restricted meanings and adds explicit premature closure and joint-decision metrics. This is a correction of evaluation semantics, not a change to the frozen detector or evidence that either failure occurred in a particular C01 record.

APPENDIX F RETROSPECTIVE RAW COMPARISON AND INDEPENDENT UNITS

The archived RAW0001 table contains 21 contexts, including nine labeled-normal and twelve abnormal contexts. Its recovered loader places a 30-s window three seconds before a supplied abnormal onset, and selects normal contexts from supplied normal intervals. Even a successful reproduction would measure classification and localization within event-guided contexts, not continuous detection. However, the recovered calendar is incompatible with the event annotations in the actual RAW files. File-history checks show that those RAW files have not changed, and the timing differences cannot be corrected by a common offset. The archived score is therefore not a reproduced comparison endpoint. The unsuc-

cessful reconstruction and its provenance audit are retained; predictions from mismatched windows are not counted as accuracy against the archived labels.

A separate complete-record replay uses fixed windows without event-guided placement and retains all normal, physical, missing and corrupt periods. Both interfaces preserve availability failures and abstentions. Physical and integrity locations are scored separately where their meanings differ; the legacy single-label export is retained for comparison. Physical-source scoring additionally requires verified source annotations; an event-code sequence alone cannot identify every intervention boundary or source. The separately sealed successor analysis uses the recovered original source workbook for physical identity, retaining current event-code support for window membership. Its coarse timing does not resolve the individual actuation or restoration times. This replay is not an official competition score. RAW0001 previously informed F01 calibration, so even a frozen new replay remains retrospective.

Simulation partitions use the complete physical parent and all of its derived windows, conditions and noise realizations. Validation belongs inside the development fraction. Repartitioning previously inspected campaigns does not restore independent-test status. Uncertainty calculations resample physical parents, retaining their paired conditions together. Different PMU channels and adjacent 30-Hz reports are not independent trials.

APPENDIX G REPOSITORY BOUNDARIES AND REPRODUCTION

The article and current supplement compile independently from `main.tex` and `supplement.tex`. The separate `historical_notes.tex` preserves corrected geometry and evidence theorems, earlier state reconstruction, the historical batch estimator, the proposed joint estimator and their original empirical qualifications. Moving this material does not remove negative results or upgrade an unexecuted estimator to an implemented one.

The submission successor is versioned in the research repository under experiments/PAPER_SUBMISSION_S01/. It contains the source provenance, observation-only interface, metric specification, synthetic regressions, and separate historical-window and complete-record outputs. Performance tables enter the manuscript only after prediction sealing and evaluation. A successful code check or PDF build is not itself a performance result.

APPENDIX H EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.

For a load ON alternative, write the innovation as $e_k - a s_k$. The affine conditional state is $m_k + a d_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{10}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{11}$$

$$C = -\frac{1}{2} \sum_k \left(e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{12}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a} U - \tfrac{1}{2} \widehat{a}^2 G - C_{ref}. \tag{13}$$

Here C_{ref} is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c) P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The H04 comparison contains no such post-startup missing observations.

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution

TABLE 1. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

APPENDIX I GROUPED LEARNING AND INTEGRITY PUBLICATION

A. RETROSPECTIVE SIM TRAINING CONTRACT

The SIM11–19 registry contains 176 observation records from 88 physical cases. Connected components join shared physical histories and observation replicas before splitting. Of 56 dependency groups, 31 are used for fitting, 8 for model selection and 17 for the reserved development evaluation: 90, 16 and 70 records, respectively. This implements the requested approximately 70/30 development/reserved partition with an internal validation subset. All campaigns were already open; the split is retrospective and does not establish unseen-source or external generalization.

Both ExtraTrees variants use 400 trees, minimum leaf size two, square-root feature sampling, balanced class weights and seed 2601026. Fitting uses every third report starting at index 360. The target is normal outside an intervention and one of 18 load/line sources during it; recovery is not a separate training class. The baseline has 846 trailing features; SHORT adds 611 features. Their input is the same 47-channel representation with a 300-report empirical reference, colored filtering and median/MAD scaling. No DAE fit or RAW annotation enters fitting. The empirical reference has no startup stationarity guarantee; it is distinct from the physical filter's admission gate.

The publication adapter requires six consecutive winning source labels. Scores are not claimed to be calibrated probabilities. Selection maximizes validation diagnoses with correct family and source emitted before OFF, then minimizes extra alarms, preferring the baseline on a complete tie. SHORT is frozen after 108/192 versus 92/192 validation successes. There is no refit on validation and no subsequent parameter or threshold selection from the reserved 70 records. The declared metric associates the first temporal alarm independently of its predicted class, within 30 seconds or before the next onset, whichever comes first; success also requires diagnosis before the current OFF.

The reserved result is 302/576, comprising 28/288 load and 274/288 line interventions, with 3 extra alarms. Only two distinct normal dependency groups exist: one in training and one reserved; validation has none. The 3696 normal seconds are repeated observations of the reserved normal group, with 0 alarms. They do not provide independent long-duration operational exposure. In particular, neither report-level binomial intervals nor a validation false-alarm rate would be justified. Load weakness and the common F01 ancestry limit transfer claims even when the group assignment is disjoint.

The selected model is also replayed on all twelve H04 records, using the same successor event metrics as the physical controls. It obtains 27/54 timely decisions, 3 extras, 3 orphan restorations and 27 missing closures, including missed events. All missed events are loads; all tested lines are diagnosed before OFF. Its six-argmax publisher has no confidence threshold, so this is not an identical decision-clock ablation. The three H04 event groups are disjoint from the SIM fitting, validation and reserved groups, but its single normal group is shared with the reserved group. The inference-only replay reads no truth, registry or split. An earlier roster contained unused event/normal metadata; it is preserved and documented. Removing that metadata reproduces every output array and event exactly on all twelve records, before separate post-seal scoring.

B. MISSING-PMU NOTICE TIMING

The continuous RAW replay contains 4 observed missing intervals. The exact submitted artifact and the immediate availability interface both identify all intervals and their affected PMU. First correct notices have mean delay 12.13 versus 0 seconds, respectively. The clock is the received report timestamp relative to the first affected report; zero means emission on that report, not zero computation or communication latency. Notices before onset cannot match a later missing interval, and a later correct source cannot repair an incorrect first notice. This is a publication benefit from already-observed flags and finite-value masks, not evidence of a new malicious-attack detector or improved physical-source inference.

C. SEGMENTED ML TRANSFER AND SOURCE COMPOSITION

The classifier consumes the phase-A voltage/current magnitudes and angles, frequency and ROCOF in the declared eight-PMU order. Complete segments require all 48 measurements finite and all presence flags true; missing reports are never removed to join their neighboring samples. Each segment gets its own 300-report empirical reference and remains unavailable through index 359. Segment startup is not certified normal and physical state is not preserved across gaps. The same frozen classifier and publication rule are used; no RAW source label, event code or score enters inference.

For a submitted-artifact prediction of line or load, including its predicted physical component in a mixed missing case, source replacement requires the first ML diagnosis of an episode with that same predicted family, emitted inside the completed window no later than the artifact's own emission. Later revisions cannot repair that first diagnosis. The earliest eligible diagnosis is selected; otherwise the ZIP source is retained. Fault, generation, integrity outputs and all window class labels and times are unchanged. The archived Top-3 list remains identified as the original ZIP list; it is not presented as the hybrid's reranked posterior.

The completed replay has 5 complete segments and 1 selected source substitution, with 179 fallbacks. The evaluator verifies full report coverage, explicit startup unavailability, causal message times and every replacement against the frozen rule before scoring. Source identity is supplied only to this later evaluator by the original metadata workbook: fault BUS39, line 23–24, generation BUS2 and load BUS7. Current event-code support determines which fixed windows contain each family; the workbook's approximate times do not supply exact action onsets or restorations. All physical-positive windows, including generation outside the learned catalog, enter the denominator. Paired joint correctness is 1/24 versus 2/24. These are dependent windows of one retrospective record; no population-level significance or official competition improvement is inferred.

APPENDIX J DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete F01 trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied. Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate F01 measurement operator is followed by its frozen innovations transform. For a geometric control, we profile nonnegative competing amplitudes. Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios exceeds one (Fig. 1). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an R11 improvement nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

APPENDIX K OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of R11.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h},$$
$$E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{14}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the frozen common angle drift, the three predicted histories are

$$\mu_i^{\mathrm{N}}(a) = y_0 + aD_i^0 + a^2 Q_i^0,$$
$$\mu_i^{\mathrm{R}}(a) = \mu_i^{\mathrm{N}}(a) + \widehat{b},$$
$$\mu_i^{\mathrm{E}}(a) = \mu_i^{\mathrm{R}}(a) + a\widehat{\eta}E_i. \tag{15}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The frozen F01 plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three frozen instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its archived instrumental noise is added.

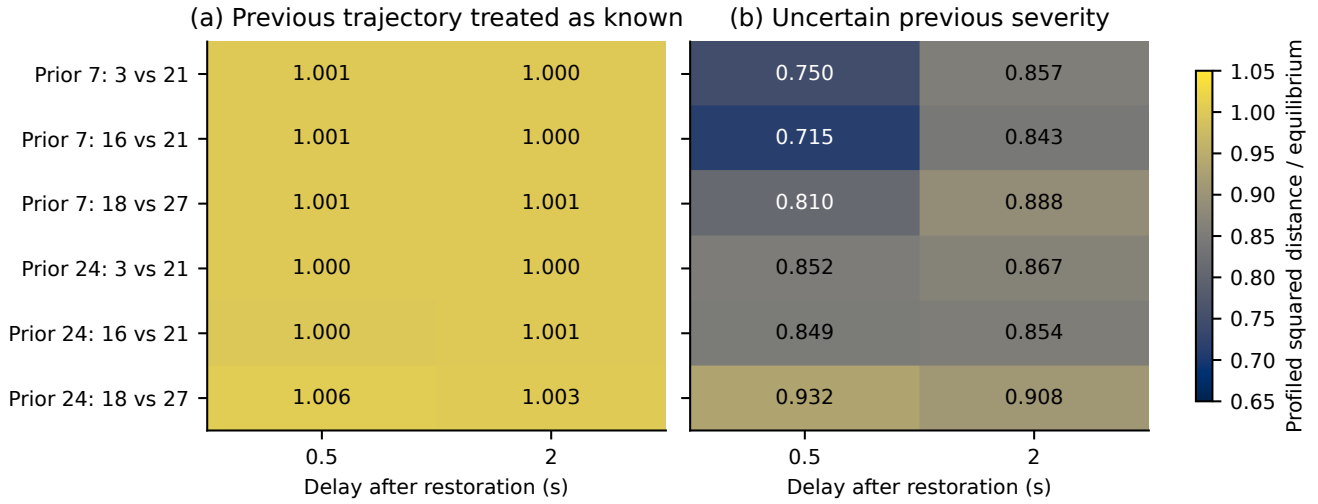


FIGURE 1. **Full-F01 physical control, not an R11 performance comparison. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

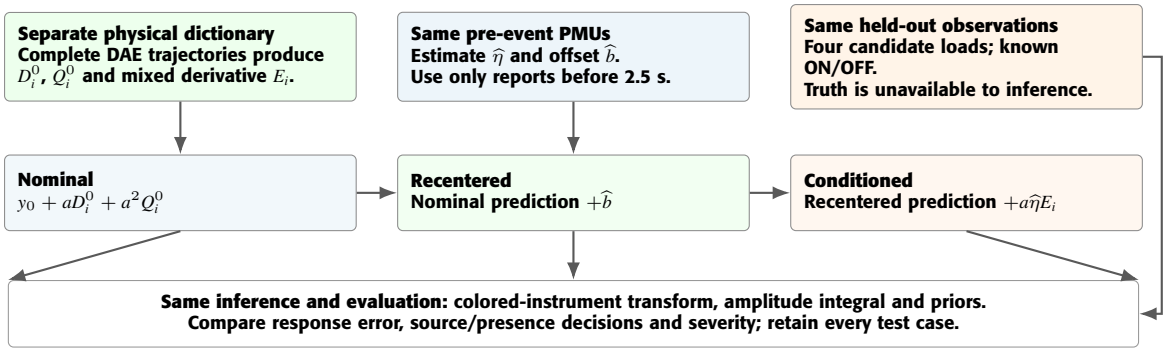


FIGURE 2. **Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.**

Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions

TABLE 2. **Controlled known-onset load test. Response error uses 80 physical cases; joint and source-only correctness use 240 noisy event records. Severity error is evaluated at the generating source. The 15 normal windows are not normal-hours exposure.**

Measure	Nominal	Recentered	Conditioned
Response error (%)	1.25	1.25	0.0561
Joint correct	114/240	240/240	240/240
Source-only correct	114/240	240/240	240/240
Normal selections	12/15	0/15	0/15
Severity error (%)	3.45	1.2	0.329

are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the

nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 3 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the frozen impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the frozen artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

APPENDIX L COMPLETE CONFIRMATORY OUTCOMES AND DIAGNOSTIC AUDIT

The frozen test contains 171 physical trajectories and 342 observed records. Every method has an accounted output for every intended record; the solver-rejection and inference-failure ledgers are empty. The continuous, persistent-line and normal physical-parent counts are 18, 27 and 12. Three network conditions and two noise levels repeat these parents. Normal controls are shared between evaluation lanes, not doubled independent exposure. Tables below are generated directly from the sealed summary, records, event matches and paired intervals.

A. TIMELY COVERAGE VERSUS CLASSIFICATION

The additional confusion view in Fig. 5 is a posthoc descriptive audit. It uses the original evaluator's temporal episode association and the first public diagnosis; it does not rematch events using source labels. A diagnosis emitted after physical OFF is counted as unavailable for this view. All diagnoses that are timely and matched in these four procedures have the correct family and exact source. The large load gap therefore reflects whether a diagnosis is available before OFF, not a measured advantage in conditional label accuracy among timely outputs. Unmatched alarms remain in the separate alarm tables. Normal records are not mixed into this event-level matrix.

B. MAINTAINING A SUSTAINED EVENT AFTER THE FIRST STOP

The first-stop score alone hides subsequent fragmentation. Although both physical banks and both learned comparators correctly identify all 162 persistent openings first, the learned comparators later return to an inactive label and issue additional alarms. Table 10 describes those already recorded outputs. It does not change the frozen first-stop outcome or extend Chen's one-stop contract into a recurrent monitor. Retained open episodes are appropriate when an intervention continues beyond the record.

C. MEASUREMENT INTEGRITY IS A SEPARATE EXECUTED COMPONENT

The availability interface correctly identifies all four missing-data intervals at the first received report that reveals the loss. This is a received-report clock, not zero transport or computation latency. Bad-data labels come from the inherited window classifier. Neither missingness nor corruption proves malicious intent. The composition preserves physical-source and affected-PMU identities but does not run a joint physical/cyber Bayesian filter; the physical bank is unavailable on the retrospective RAW0001 replay under its startup noise-scale contract.

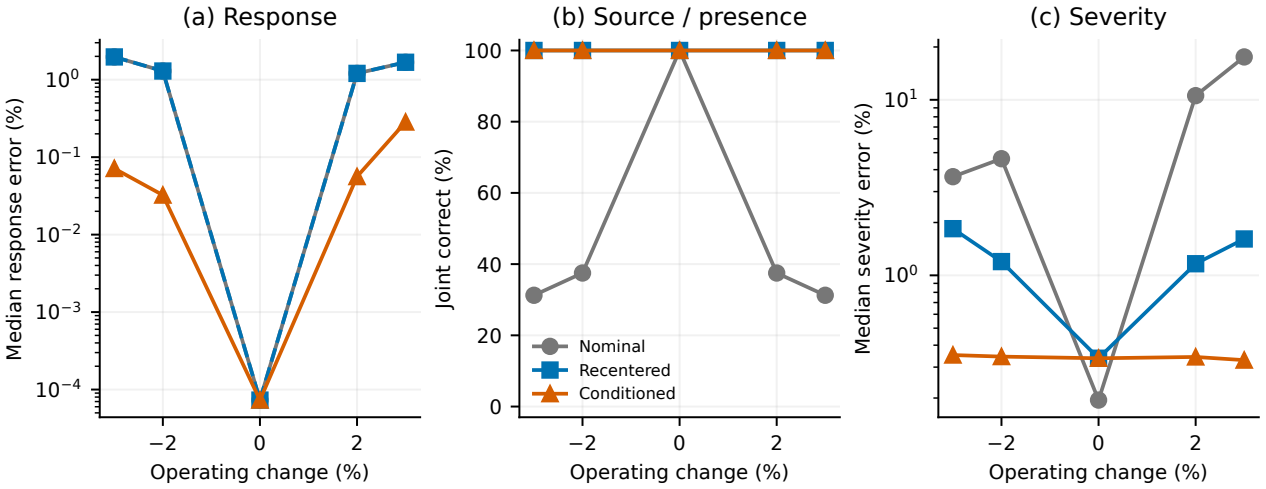


FIGURE 3. **Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.**

TABLE 3. **Continuous outcomes by network and noise stratum. Each cell is timely family/source decisions / intended events; unmatched event-stream alarms. Replays share physical parents.**

Network	Noise	Physical bank	Nominal bank	ExtraTrees	ET short
Matched	1	108/108; 1	108/108; 1	56/108; 6	60/108; 5
Matched	2	108/108; 0	108/108; 0	54/108; 4	58/108; 0
In-bank	1	108/108; 2	108/108; 3	56/108; 5	60/108; 5
In-bank	2	108/108; 1	108/108; 2	54/108; 4	57/108; 0
Out-of-bank	1	106/108; 38	105/108; 47	56/108; 5	60/108; 6
Out-of-bank	2	106/108; 30	106/108; 39	54/108; 5	57/108; 0

TABLE 4. **Continuous monitoring: unavailable intended records, normal-record alarm incidence, normal alarm counts and available normal exposure, false restorations and open episodes. Counts combine six repeated strata descriptively; normal parents are not independent across strata. Missing outputs remain intended failures.**

Method	Unavailable	Normal incidence	Normal alarms	Exposure (h)	False OFF	Open at end
Physical 5-model	0	0/72	0	11.760	58	0
Physical nominal	0	0/72	0	11.760	68	0
ExtraTrees	0	0/72	0	11.760	29	0
ET short	0	0/72	0	11.760	16	0

TABLE 5. **Conditional median diagnosis/restoration delays (s), kept within each paired stratum. These medians summarize matched emitted diagnoses/restoration messages with finite delay and must be read alongside omissions, unavailable records and censoring; – denotes no finite estimate. Logical delays do not measure computational throughput.**

R/X condition	Noise	Physical 5-model	Physical nominal	ExtraTrees	ET short
Matched	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
Matched	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
In-bank	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
In-bank	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
Out-of-bank	1	0.33/0.18	0.33/0.18	0.23/0.18	0.19/0.18
Out-of-bank	2	0.26/0.18	0.33/0.18	0.30/0.18	0.20/0.18

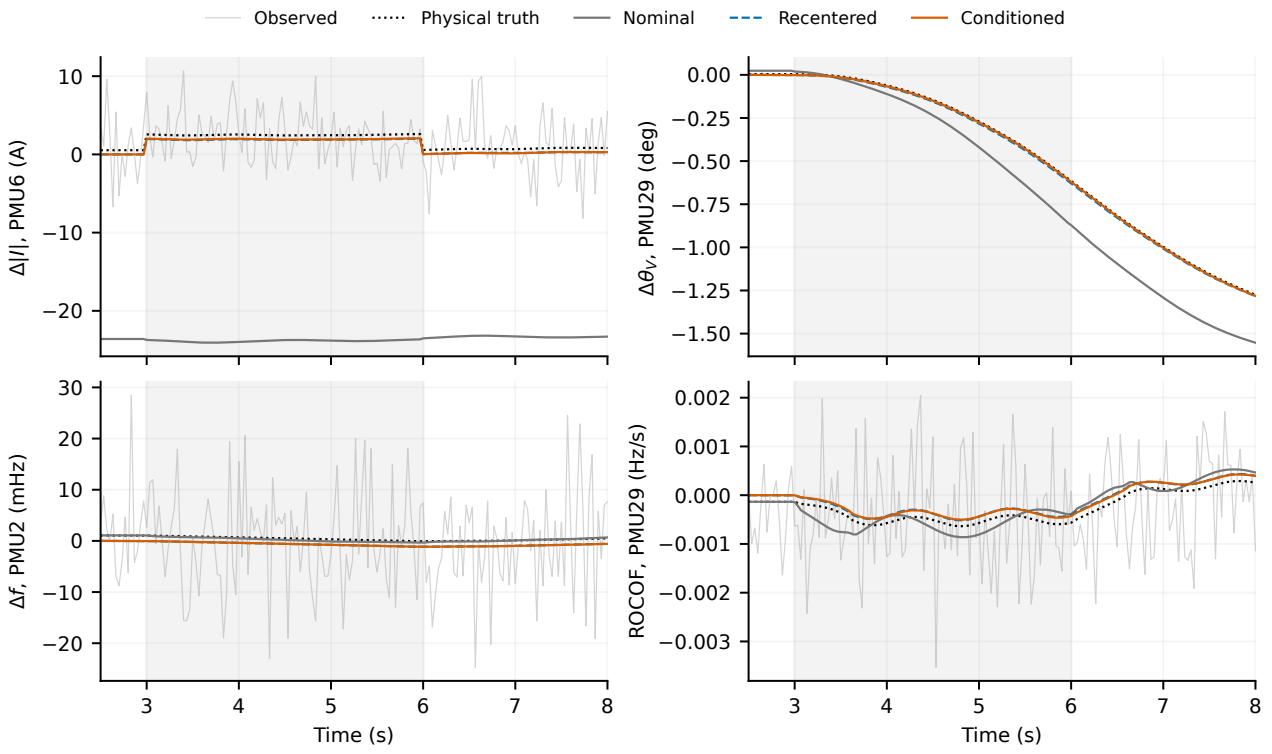


FIGURE 4. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

TABLE 6. **Persistent-line comparison on the common observations. Each cell gives correct first source / intended openings; pre-change stops / right-censored openings; mean restricted correct delay (s). Wrong first sources remain in the denominator. Restricted delay retains the horizon penalty for failures and censoring defined by the frozen evaluator.**

R/X condition	Noise	Physical 5-model	Physical nominal	Chen2016
Matched	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Matched	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
In-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.04
In-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.05
Out-of-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Out-of-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.59

TABLE 7. **Persistent-line coverage and stopping outcomes across six repeated strata, including both frozen ExtraTrees comparators. Normal controls are the same controls used in the continuous table, not additional exposure. Chen2016 normal exposure ends at its first stop; the other methods monitor the full available record.**

Method	Unavailable	Correct first	Pre-change	Censored	Normal incidence	Exposure (h)
Physical 5-model	0	162/162	0	0	0/72	11.760
Physical nominal	0	162/162	0	0	0/72	11.760
ExtraTrees	0	162/162	0	0	0/72	11.760
ET short	0	162/162	0	0	0/72	11.760
Chen2016	0	74/162	66	0	72/72	0.701

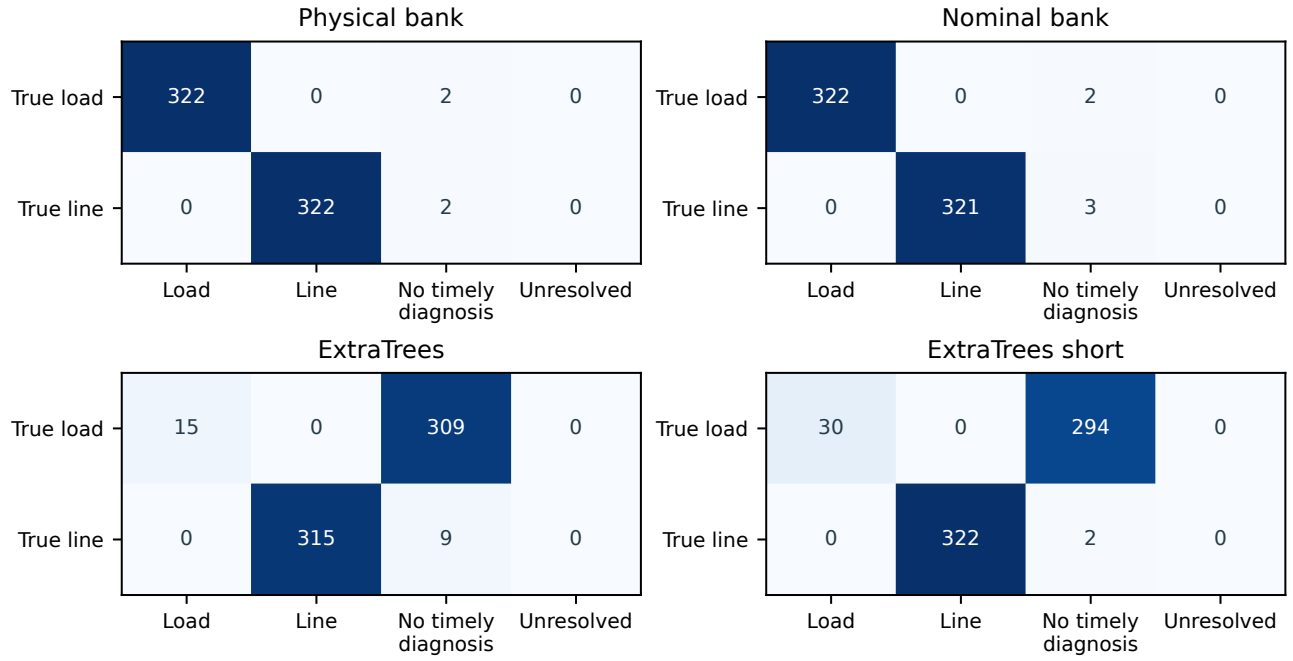


FIGURE 5. **Posthoc confusion audit of the frozen continuous test, with 324 realizations per true family. No timely diagnosis includes omitted and late diagnoses. Counts reuse the registered episode matches; all 432 prediction JSON files used were checked against the prediction seal. These panels do not include unmatched alarms or constitute a new predeclared endpoint.**

TABLE 8. **Timely decisions by source in the continuous test. These are correlated realizations of the registered physical schedules, not source-excluded generalization.**

Family / asset	Physical bank	Nominal bank	ExtraTrees	ET short
Line / LINE14-15	36/36	36/36	27/36	34/36
Line / LINE16-17	36/36	36/36	36/36	36/36
Line / LINE17-18	48/48	48/48	48/48	48/48
Line / LINE23-24	30/30	30/30	30/30	30/30
Line / LINE26-27	24/24	24/24	24/24	24/24
Line / LINE26-29	24/24	23/24	24/24	24/24
Line / LINE3-4	42/42	40/42	42/42	42/42
Line / LINE4-14	46/48	48/48	48/48	48/48
Line / LINE9-39	36/36	36/36	36/36	36/36
Load / BUS15	42/42	42/42	6/42	6/42
Load / BUS16	42/42	42/42	0/42	3/42
Load / BUS20	12/12	12/12	6/12	12/12
Load / BUS24	42/42	42/42	0/42	0/42
Load / BUS26	18/18	18/18	0/18	0/18
Load / BUS3	42/42	42/42	3/42	3/42
Load / BUS31	40/42	40/42	0/42	0/42
Load / BUS4	36/36	36/36	0/36	6/36
Load / BUS7	48/48	48/48	0/48	0/48

TABLE 9. Paired physical-bank minus learned-comparator differences. Timely attribution is in percentage points; extras are per full event-record hour, including startup. Brackets are the frozen 95-percent parent-bootstrap intervals.

Network	Noise	Method	Reference	Timely difference	Extra-rate difference
Matched	1	Physical bank	ExtraTrees short	44.44 [40.48, 47.92]	-6.67 [-15.00, 0.00]
Matched	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-8.33 [-18.33, 0.00]
Matched	1	Nominal bank	ExtraTrees short	44.44 [40.74, 48.04]	-6.67 [-13.33, 0.00]
Matched	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-8.33 [-18.33, 0.00]
Matched	2	Physical bank	ExtraTrees short	46.30 [42.75, 49.12]	0.00 [0.00, 0.00]
Matched	2	Physical bank	ExtraTrees	50.00 [46.30, 53.70]	-6.67 [-13.33, -1.67]
Matched	2	Nominal bank	ExtraTrees short	46.30 [42.86, 49.12]	0.00 [0.00, 0.00]
Matched	2	Nominal bank	ExtraTrees	50.00 [46.38, 53.70]	-6.67 [-13.33, -1.67]
In-bank	1	Physical bank	ExtraTrees short	44.44 [40.62, 47.92]	-5.00 [-13.33, 5.00]
In-bank	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-5.00 [-13.33, 5.00]
In-bank	1	Nominal bank	ExtraTrees short	44.44 [40.62, 47.92]	-3.33 [-11.67, 6.67]
In-bank	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-3.33 [-11.67, 5.00]
In-bank	2	Physical bank	ExtraTrees short	47.22 [43.18, 50.98]	1.67 [0.00, 5.00]
In-bank	2	Physical bank	ExtraTrees	50.00 [46.49, 53.70]	-5.00 [-10.00, 0.00]
In-bank	2	Nominal bank	ExtraTrees short	47.22 [43.33, 51.04]	3.33 [0.00, 8.33]
In-bank	2	Nominal bank	ExtraTrees	50.00 [46.30, 53.70]	-3.33 [-10.00, 3.33]
Out-of-bank	1	Physical bank	ExtraTrees short	42.59 [38.60, 46.43]	53.33 [36.67, 71.67]
Out-of-bank	1	Physical bank	ExtraTrees	46.30 [42.11, 50.00]	55.00 [36.67, 73.33]
Out-of-bank	1	Nominal bank	ExtraTrees short	41.67 [36.11, 46.67]	68.33 [46.67, 90.00]
Out-of-bank	1	Nominal bank	ExtraTrees	45.37 [41.11, 50.00]	70.00 [45.00, 95.00]
Out-of-bank	2	Physical bank	ExtraTrees short	45.37 [41.11, 50.00]	50.00 [35.00, 65.00]
Out-of-bank	2	Physical bank	ExtraTrees	48.15 [43.59, 52.50]	41.67 [25.00, 60.00]
Out-of-bank	2	Nominal bank	ExtraTrees short	45.37 [41.11, 50.00]	65.00 [41.67, 88.33]
Out-of-bank	2	Nominal bank	ExtraTrees	48.15 [43.75, 52.56]	56.67 [31.67, 80.00]

TABLE 10. Descriptive persistence audit after first detection on sustained line openings. All four procedures first identify 162/162 lines. Extras are later unmatched alarms; unmatched OFF counts only restoration messages on unmatched episodes. Open-at-end messages are expected for sustained events and are not automatically errors. False-negative reports include detection delay and later inactive labels. Chen is a first-stop method and is excluded from this recurrent-monitoring diagnostic.

Method	Openings	Extras	Unmatched OFF	Open at end	FN reports
Physical bank	162	0	0	162	810
Nominal bank	162	0	0	162	810
ExtraTrees	162	164	164	16	335374
ExtraTrees short	162	179	178	61	182600

TABLE 11. Retrospective measurement-integrity components over 180 windows. Hits require family and PMU identity. Missingness is observed availability; corrupted-data identification is inherited from the window classifier. These are not counts of attributed cyberattacks.

Component	Positive-window hits	FP	FN
Missing	14/14	0	0
Corrupt	1/2	2	1

APPENDIX M CONTROLLED COMPONENT STUDY: PENDING OUTPUTS

The separate factorial run must account for all 24 records under all four arms before aggregation. Its results are not yet available in this draft. The original test protocol, predictions and metrics are unchanged.

REFERENCES

[1] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.